

Centered Natural-Parameter Group Selection for Posterior-Score Contrast Support in von Mises–Fisher Mixtures

Hyunsoo Shin^a, Byungtae Seo^{a,*}

^a*Department of Statistics, Sungkyunkwan University, Seoul 03063, Republic of Korea*

Abstract

Keywords: Directional data; mixture model; von Mises–Fisher distribution; heterogeneity selection; group lasso; posterior-score contrast support

1. Introduction

Directional observations include normalized documents, compositional profiles, and hyperspherical embeddings. Finite von Mises–Fisher (vMF) mixtures provide a model-based alternative to spherical k -means (Banerjee et al., 2005), but high-dimensional fits can be difficult to estimate and interpret.

Sparse vMF mixtures have primarily penalized component directions to estimate sparse prototypes (Rossi and Barbaro, 2022). Prototype presence is not the same as contribution to posterior component comparisons: a coordinate can be active in every prototype yet cancel from score contrasts, whereas unequal concentrations can amplify modest directional differences.

Related mixture methods use entrywise shrinkage, grouped means, pairwise fusion, or discriminative subspaces (Pan and Shen, 2007; Xie et al., 2008; Guo et al., 2010; Bouveyron and Brunet, 2012; Bouveyron and Brunet-Saumard, 2014); sum-to-zero contrasts separate common levels from heterogeneity (Bondell and Reich, 2009). Directional-mixture alternatives target saliency, blocks, or hierarchical regularization (Amayri and Bouguila, 2013; Salah et al., 2016; Gopal and Yang, 2014; Taghia et al., 2014; Salah

*Corresponding author.

Email address: seobt@skku.edu (Byungtae Seo)

and Nadif, 2019). Here component-specific concentration makes natural-parameter, rather than directional, contrasts relevant to posterior scores.

Separating support construction from likelihood refitting parallels Lasso–MLE model collections for Gaussian mixtures (Meynet, 2012; Meynet and Maugis-Rabusseau, 2012). Our construction instead aligns both stages with centered vMF natural-parameter support, which equals the support of pair-wise posterior-score linear coefficients. The common baseline cancels from those coefficients but remains in the fitted density.

This paper targets posterior-score contrast support through natural parameters formed by multiplying each component direction by its concentration. For each observed coordinate, the component-specific natural parameters are decomposed into a common baseline and a centered contrast. The proposed centered natural-parameter group lasso (E-CGL) penalty groups the centered contrast across components, so its unit of regularization is the same coordinate-level unit used to define the target support. The common baseline remains unpenalized. An adaptive centered natural-parameter group lasso (E-ACGL) is retained as a secondary extension; E-CGL remains the primary estimator.

Candidate supports are generated by a guarded path with proximal-gradient updates inside a generalized expectation-maximization (GEM) algorithm. They are compared after support-constrained refitting that preserves the target. The primary selector is a practical Bayesian information criterion (BIC) computed from the refitted likelihood; alternative degrees of freedom and extended BIC (EBIC) rules are treated as sensitivity analyses. Support-recovery claims condition on a fixed number of components. When that number is unknown, component-number selection and support selection are examined in separate stages rather than through a single joint criterion.

The empirical study has three parts. Oracle-error calibrated simulations compare posterior-score contrast support recovery across sample size, ambient dimension, and concentration heterogeneity. Shared-natural-parameter and dense-support designs separate the proposed estimand from prototype sparsity and identify settings in which assuming sparse posterior-score contrast support is inappropriate. A full-data Classic3 analysis examines clustering, coordinate reduction, and token-level contrasts on a vocabulary-aligned directional representation. Prespecified Classic3 splits assess support stability.

The contributions are therefore as follows: (i) a posterior-score contrast support estimand for vMF mixtures, its centered representation, and its de-

composition into directional and concentration effects; (ii) E-CGL, together with guarded estimation and target-preserving refitting, with E-ACGL as an adaptive extension using fixed pilot weights; (iii) existence on the bounded computational parameter space, convergence of the fixed-responsibility proximal algorithm, and conditional stationarity of accumulation points under stated regularity conditions; and (iv) an empirical evaluation that documents sparse, prototype, and dense-support regimes. E-CGL is not intended to replace sparse-prototype methods when prototype sparsity is the scientific target.

The remainder of the paper is organized as follows. Section 2 presents the vMF model, posterior-score contrast support, and centered group regularization. Section 3 develops the guarded proximal GEM estimator, pathwise support selection and refitting, and the structural and algorithmic properties used in the analysis. Sections 4 and 5 report the numerical studies and real-data analyses, respectively; the latter use prespecified preprocessing rules and split indices. Section 6 summarizes the main findings, limitations, and conclusions.

2. Model and centered regularization

2.1. vMF mixture in natural parameters

Let $x_i \in \mathbb{S}^{d-1}$, $i = 1, \dots, n$, denote unit-normalized directional observations. A K -component vMF mixture has density

$$f(x_i; \Theta) = \sum_{k=1}^K \pi_k C_d(\kappa_k) \exp\{\kappa_k \mu_k^\top x_i\},$$

where $\pi_k > 0$, $\sum_k \pi_k = 1$, $\|\mu_k\|_2 = 1$, and $\kappa_k \geq 0$. Its normalizing constant is

$$C_d(\kappa) = \frac{\kappa^{d/2-1}}{(2\pi)^{d/2} I_{d/2-1}(\kappa)}.$$

Here I_ν is the modified Bessel function of the first kind of order ν .

At $\kappa = 0$, this expression is interpreted by continuous extension as $C_d(0) = \Gamma(d/2)/(2\pi^{d/2})$.

Write

$$\begin{aligned} \Theta &= (\pi_1, \dots, \pi_K, \eta_1, \dots, \eta_K), \\ E &= (\eta_1, \dots, \eta_K)^\top, & M &= (\mu_1, \dots, \mu_K)^\top, \end{aligned}$$

where $E, M \in \mathbb{R}^{K \times d}$ and $\eta_k = \kappa_k \mu_k \in \mathbb{R}^d$. Then

$$f(x_i; \Theta) = \sum_{k=1}^K \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i).$$

The component-level mapping back to (μ_k, κ_k) and its zero-concentration exception are recorded in Section 3.4. The observed log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^n \log \left\{ \sum_{k=1}^K \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i) \right\},$$

where the component densities are defined with respect to surface-area measure on $\mathbb{S}^{d-1}$.

2.2. Posterior-score contrast support

Let $Z \in \{1, \dots, K\}$ denote the latent component label for a generic observation x . The component score

$$s_k(x) = \log \pi_k + \log C_d(\|\eta_k\|_2) + \eta_k^\top x$$

implies the pairwise posterior log-odds representation

$$\log \frac{\Pr(Z = k \mid x)}{\Pr(Z = \ell \mid x)} = b_{k\ell} + (\eta_k - \eta_\ell)^\top x,$$

$$b_{k\ell} = \log \frac{\pi_k}{\pi_\ell} + \log \frac{C_d(\|\eta_k\|_2)}{C_d(\|\eta_\ell\|_2)}.$$

Thus $\eta_k - \eta_\ell$, rather than $\mu_k - \mu_\ell$, is the coefficient of the feature-dependent linear term. Let $\eta_{\cdot j}$ denote column j of E , and define

$$H_K = I_K - K^{-1} \mathbf{1}_K \mathbf{1}_K^\top, \quad c_{\cdot j} = H_K \eta_{\cdot j} = \eta_{\cdot j} - \bar{\eta}_j \mathbf{1}_K,$$

where $\bar{\eta}_j = K^{-1} \sum_k \eta_{kj}$. We define

$$S_\eta = \{j : \|c_{\cdot j}\|_2 > 0\}.$$

For population natural parameters $E^\star$, write $S_\eta^\star = \{j : \|H_K E_{\cdot j}^\star\|_2 > 0\}$.

We use *posterior-score contrast support* for this feature-dependent linear support. It does not include the intercepts $b_{k\ell}$. A coordinate with $c_{\cdot j} = 0$ cancels from every pairwise linear term, although its common baseline $\bar{\eta}_j$ remains in the fitted component densities and may affect $b_{k\ell}$ through the

concentration norms. Proposition 1 shows that this support is the union of the feature-dependent linear supports in all pairwise posterior log-odds and gives the corresponding pairwise-dispersion identity.

For directional-support comparisons, assume $\kappa_k > 0$ for every component, so that $\mu_k = \eta_k / \|\eta_k\|_2$ is identified. Then let

$$\bar{\mu}_j = K^{-1} \sum_{k=1}^K \mu_{kj}, \quad c_{\cdot j}^{(\mu)} = H_K M_{\cdot j} = M_{\cdot j} - \bar{\mu}_j \mathbf{1}_K,$$

and define

$$S_P = \{j : \|M_{\cdot j}\|_2 > 0\}, \quad S_\mu = \{j : \|c_{\cdot j}^{(\mu)}\|_2 > 0\}.$$

The set S_P records coordinates present in at least one directional prototype, whereas S_μ records between-component variation in mean directions. The proposed S_η records coordinates entering posterior-score contrasts. The three targets can differ when component concentrations are heterogeneous. If any $\kappa_k = 0$, the corresponding direction is unidentified and S_P and S_μ require an explicit directional convention; S_η remains well-defined through the natural parameters.

Accordingly, $c_{\cdot j} \neq 0$ defines a posterior-score contrast coordinate; $c_{\cdot j} = 0$ with $\bar{\eta}_j \neq 0$ defines a common coordinate; and $c_{\cdot j} = 0$ with $\bar{\eta}_j = 0$ defines a globally null coordinate. These categories distinguish the proposed target from prototype support, which records nonzero component coordinates without requiring a difference between components.

The support is invariant to permutations of component labels because such a permutation only reorders the entries of $c_{\cdot j}$. It is not invariant to an arbitrary orthogonal rotation of the feature axes. As with other coordinate-selection procedures, its interpretation is therefore relative to the chosen coordinate system.

2.3. Centered natural-parameter group lasso (E-CGL)

The proposed E-CGL method regularizes the vMF natural parameters $\eta_k = \kappa_k \mu_k$, collected row-wise in $E \in \mathbb{R}^{K \times d}$. It penalizes the coordinate-wise contrasts $H_K E_{\cdot j}$ while leaving the common natural-parameter baseline unpenalized. For fixed K , the primary penalty functional is

$$\mathcal{P}_{\text{CGL}}(E) = \sum_{j=1}^d \|H_K E_{\cdot j}\|_2 = \sum_{j=1}^d \|c_{\cdot j}\|_2.$$

Each group is one observed coordinate across the K component contrasts, so the unit of regularization matches the definition of S_η . This coordinate-group construction follows the grouped-variable regularization principle of Yuan and Lin (2006). By contrast,

$$\|\eta_j\|_2^2 = K\bar{\eta}_j^2 + \|c_j\|_2^2,$$

so a raw- η group penalty combines common and contrast effects. A centered direction-space penalty targets S_μ , whereas the Rossi-style entrywise penalty targets S_P . When the κ_k differ, neither target is generally equal to the coefficient support of $\eta_k - \eta_\ell$ in posterior log-odds. These are distinct estimands, not competing parameterizations of the same support.

An entrywise centered penalty selects component-coordinate entries separately: $\lambda_\eta \sum_{k,j} |c_{kj}|$. E-CGL instead selects the complete coordinate contrast as one group. The resulting proximal map and label invariance are stated in Propositions 3 and 4.

2.4. Adaptive weighted extension (E-ACGL)

E-ACGL applies adaptive penalization at the coordinate-group level using weights computed once from a dense pilot, following adaptive lasso and adaptive group-lasso principles (Zou, 2006; Wang and Leng, 2008). Let $a \in \mathcal{A} = \{\text{CGL}, \text{ACGL}\}$ denote the method and let E^D be the retained dense pilot. Define

$$w_j^{(a)} = \begin{cases} 1, & a = \text{CGL}, \\ \frac{\left(\|H_K E_{\cdot j}^D\|_2 + \epsilon\right)^{-\gamma}}{\text{median}_{1 \leq h \leq d} \left(\|H_K E_{\cdot h}^D\|_2 + \epsilon\right)^{-\gamma}}, & a = \text{ACGL}, \end{cases} \quad (1)$$

$$\mathcal{P}_a(E) = \sum_{j=1}^d w_j^{(a)} \|H_K E_{\cdot j}\|_2.$$

The two methods share the dense pilot. E-CGL, the primary estimator, is the special case $w_j^{(a)} = 1$ for every j . E-ACGL is an adaptive extension with fixed pilot weights, using $\gamma = 1$ and $\epsilon = 10^{-6}$. Both procedures target S_η . For finite positive weights, the results below extend by replacing λ_η with $\lambda_\eta w_j^{(a)}$ in group j . No adaptive oracle claim is made.

3. Estimation by guarded proximal generalized EM

3.1. Penalized objective

The finite vMF mixture likelihood can diverge under concentration collapse (Ng, 2023). We therefore define the fitted model on

$$\mathcal{T}_{\kappa_{\max}} = \left\{ (\pi, E) : \pi \in [0, 1]^K, \mathbf{1}_K^\top \pi = 1, \|\eta_k\|_2 \leq \kappa_{\max} \text{ for every } k \right\},$$

with $\kappa_{\max} = 10^6$ in all reported analyses. The numerical mixing-weight floor is an implementation safeguard; the existence argument below uses the closed simplex, and the local convergence statement assumes the floor and the concentration bound are inactive.

For fixed K , λ_η , and $a \in \mathcal{A}$, the path-generating criterion is

$$\mathcal{L}_{\lambda_\eta}^{(a)}(\Theta) = \ell(\Theta) - \lambda_\eta \mathcal{P}_a(E). \quad (2)$$

The penalty and the vMF normalizing constant are coupled through η_k , so the natural-parameter update is computed by a guarded proximal-gradient generalized M-step.

Lemma 1 (Existence on the finite parameter space). *For fixed K, d, n , finite $\kappa_{\max}$, nonnegative λ_η , and finite positive fixed weights, the criterion in (2) attains a maximum on $\mathcal{T}_{\kappa_{\max}}$.*

The lemma establishes existence for the bounded computational problem. It does not imply uniqueness, mixture identifiability, or attainment of a global maximum by the numerical algorithm.

3.2. Fixed-penalty guarded proximal GEM

We use a generalized expectation-maximization (GEM) procedure at each fixed value of λ_η . The E-step computes the posterior responsibilities exactly. In the generalized M-step, the mixing proportions are updated by (8), while the natural-parameter block receives finitely many safeguarded proximal-gradient updates. This block is required to increase the penalized auxiliary criterion within numerical tolerance rather than to attain its exact conditional maximum. The procedure is therefore a GEM algorithm rather than an exact EM algorithm.

3.2.1. *E-step and fixed-responsibility subproblem*

At outer iteration t , compute $\tau_{ik}^{(t)}$ and

$$\tau_{ik}^{(t)} = \frac{\pi_k^{(t)} C_d(\|\eta_k^{(t)}\|_2) \exp\{(\eta_k^{(t)})^\top x_i\}}{\sum_{h=1}^K \pi_h^{(t)} C_d(\|\eta_h^{(t)}\|_2) \exp\{(\eta_h^{(t)})^\top x_i\}}, \quad (3)$$

and define

$$N_k^{(t)} = \sum_{i=1}^n \tau_{ik}^{(t)}, \quad r_k^{(t)} = \sum_{i=1}^n \tau_{ik}^{(t)} x_i.$$

Conditional on these responsibilities and the fixed weights for method a , the cap-inactive natural-parameter block is represented by

$$\begin{aligned} \min_E \quad & F_t^{(a)}(E) = f_t(E) + g_{\lambda_\eta}^{(a)}(E), \\ & f_t(E) = - \sum_{k=1}^K \left\{ (r_k^{(t)})^\top \eta_k + N_k^{(t)} \log C_d(\|\eta_k\|_2) \right\}, \\ & g_{\lambda_\eta}^{(a)}(E) = \lambda_\eta \sum_{j=1}^d w_j^{(a)} \|H_K \eta_j\|_2. \end{aligned} \quad (4)$$

The numerical algorithm evaluates this objective through cap-feasible trial iterates. It rejects rather than projects a proposal that violates the concentration cap; the cap-active constrained subproblem is not claimed to be solved by the closed-form proximal update.

The smooth term is convex because $-\log C_d(\|\eta_k\|_2)$ is the vMF log-partition function. Its gradient is

$$\nabla_{\eta_k} f_t(E) = N_k^{(t)} A_d(\|\eta_k\|_2) \frac{\eta_k}{\|\eta_k\|_2} - r_k^{(t)}, \quad (5)$$

with the continuous value $-r_k^{(t)}$ at $\eta_k = 0$, where

$$A_d(\kappa) = \frac{I_{d/2}(\kappa)}{I_{d/2-1}(\kappa)}$$

is the vMF mean-resultant-length function, with $A_d(0) = 0$ by continuous extension.

3.2.2. Generalized M-step and centered proximal update

Let $J_K = K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For a step size $s > 0$, set

$$V = E^{(u)} - s\nabla f_t(E^{(u)}). \quad (6)$$

The update separates by observed coordinate and has the closed form

$$E_{\cdot j}^{(u+1)} = J_K V_{\cdot j} + \left(1 - \frac{s\lambda_\eta w_j^{(a)}}{\|H_K V_{\cdot j}\|_2}\right)_+ H_K V_{\cdot j}. \quad (7)$$

Here $(z)_+ = \max(z, 0)$, and the second term is defined as zero when $H_K V_{\cdot j} = 0$.

The common baseline is therefore retained, whereas the centered contrast is group-thresholded; this is a proximal group soft-thresholding operation (Parikh and Boyd, 2014). Mixing proportions are updated by

$$\pi_k^{\text{cand}} = \frac{\max(N_k^{(t)}/n, 10^{-12})}{\sum_h \max(N_h^{(t)}/n, 10^{-12})}. \quad (8)$$

The initial step size is $s = 1/\max_k N_k^{(t)}$. Define the majorization residual

$$R_{\text{maj}}(E^+, E; s) = f_t(E^+) - f_t(E) - \langle \nabla f_t(E), E^+ - E \rangle - \frac{\|E^+ - E\|_F^2}{2s}. \quad (9)$$

The smooth majorization condition is equivalent to $R_{\text{maj}}(E^+, E; s) \leq 0$. Numerically, the update is accepted only when $R_{\text{maj}}(E^+, E; s) \leq \varepsilon_{\text{maj}}$ and $\max_k \|\eta_k^+\|_2 \leq \kappa_{\text{max}}$. Otherwise, the proposal is not projected in this proximal loop; the step is halved and the proposal is recomputed. If no acceptable proposal is found in 40 trials, the current path point is marked invalid. The path driver skips that tuning value, retries the next value from the last accepted fit, and terminates the remaining path only after three consecutive failures.

3.2.3. Safeguards and stopping rules

Let $Q(\Theta \mid \tau)$ be the expected complete-data log-likelihood, using the standard convention $0 \log 0 = 0$ for zero-weight terms. For an old iterate Θ and candidate Θ^+ , define

$$\begin{aligned}
Q_{\lambda_\eta}^{(a)}(\Theta \mid \tau) &= Q(\Theta \mid \tau) - \lambda_\eta \mathcal{P}_a(E), \\
\Delta Q_{\lambda_\eta}^{(a)} &= Q_{\lambda_\eta}^{(a)}(\Theta^+ \mid \tau) - Q_{\lambda_\eta}^{(a)}(\Theta \mid \tau), \\
\Delta \mathcal{L}_{\lambda_\eta}^{(a)} &= \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^+) - \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta).
\end{aligned} \tag{10}$$

The guarded generalized M-step accepts the candidate only if

$$\Delta Q_{\lambda_\eta}^{(a)} \geq -\varepsilon_{\text{acc}}, \quad \Delta \mathcal{L}_{\lambda_\eta}^{(a)} \geq -\varepsilon_{\text{acc}}. \tag{11}$$

Otherwise, the path fit returns only the previous accepted estimate, accompanied by a failure diagnostic; its support is not recorded, and the path driver proceeds to the next tuning value from the last accepted fit. Three consecutive failed tuning values terminate the remaining path.

Components with an effective count below 10^{-8} are retained to preserve the fixed value of K , and this event is recorded. They are not pruned or reinitialized. These safeguards support statements about accepted numerical updates; they do not establish convergence of the complete parameter sequence, stationarity of every fitted path point, or global optimality.

The recorded analyses use the absolute support threshold $\varepsilon_{\text{supp}} = 10^{-8}$. This threshold is a numerical convention for reading a fitted path support, not an additional penalty parameter.

The recorded simulation and real-data controls use at most 200 and 160 proximal-gradient iterations per generalized M-step, respectively. The inner and outer stopping rules are

$$\frac{\|E^{(u+1)} - E^{(u)}\|_F}{\max\{1, \|E^{(u)}\|_F\}} < \varepsilon_{\text{PG}}, \tag{12}$$

$$\frac{|\mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t+1)}) - \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t)})|}{\max\{1, |\mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t)})|\}} < \varepsilon_{\text{conv}}. \tag{13}$$

The numerical tolerances are

$$(\varepsilon_{\text{PG}}, \varepsilon_{\text{maj}}, \varepsilon_{\text{acc}}, \varepsilon_{\text{conv}}, \varepsilon_{\text{supp}}) = (10^{-8}, 10^{-10}, 10^{-8}, 10^{-7}, 10^{-8}).$$

The implementation permits at most 40 backtracking attempts and initially allows 100 outer GEM iterations at each path point. If the provisional refit-BIC winner is initialized by an unconverged penalized fit, the complete path is rebuilt with 440 additional iterations per point and only converged penalized sources are admitted to the repeated refit-BIC comparison.

3.3. Pathwise selection and support-constrained refit

3.3.1. Method-specific zero-augmented Karush–Kuhn–Tucker-scaled path

The nonfailed multistart fit with the largest finite observed log-likelihood is the provisional dense pilot. If it reaches its iteration limit, its terminal iterate is continued once for the same iteration budget. If continuation fails or remains unconverged, the procedure falls back to the largest-likelihood converged original start when available; otherwise the better finite terminal fit is retained with a fallback flag. Denote the result by $\hat{\Theta}^D$. After the weights in (1) are computed and held fixed, the implementation first obtains the guarded zero-penalty fit

$$\hat{\Theta}_0^{(a)} = \text{Fit}_{\lambda_\eta=0}^{(a)} \left(\hat{\Theta}^D; w^{(a)} \right).$$

Responsibilities τ^0 under $\hat{\Theta}_0^{(a)}$ define (N_k^0, r_k^0) for path construction. Let $r_\bullet = \sum_k r_k^0$, $N_\bullet = \sum_k N_k^0 = n$, and $\bar{\rho}_\bullet = \|r_\bullet\|_2 / N_\bullet$. The Banerjee approximation (Banerjee et al., 2005, Eq. (4.4)) is

$$\hat{\kappa}_B(\rho) = \frac{\rho(d - \rho^2)}{1 - \rho^2} \quad (14)$$

Let $\tilde{\rho}_\bullet$ denote the numerically clipped version of $\bar{\rho}_\bullet$. The pooled natural parameter used for path construction is

$$\eta^{\text{common}} = \begin{cases} \min\{\kappa_{\max}, \hat{\kappa}_B(\tilde{\rho}_\bullet)\} r_\bullet / \|r_\bullet\|_2, & \|r_\bullet\|_2 > 0, \\ 0, & \|r_\bullet\|_2 = 0. \end{cases}$$

The exact clipping constants and zero-resultant tolerance are given in the Supplementary Material. Define $E_0^{\text{com}} = \mathbf{1}_K(\eta^{\text{common}})^\top$ and $G_0 = \nabla f_{\tau^0}(E_0^{\text{com}})$, where f_{τ^0} denotes the smooth term f_t in Eq. (4) evaluated at the responsibilities τ^0 . A Karush–Kuhn–Tucker (KKT)-motivated upper path scale evaluated at this approximate pooled fit is

$$\tilde{\lambda}_{\max, \eta}^{(a)} = 1.05 \max_{1 \leq j \leq d} \frac{\|(H_K G_0)_{\cdot j}\|_2}{w_j^{(a)}}. \quad (15)$$

Conditional on E_0^{com} , the uninflated maximum in Eq. (15) is the group threshold required by the centered-block KKT condition. Because the pooled concentration is the capped Banerjee approximation rather than the numerical solution of the pooled mean-resultant equation, however, Eq. (15) is not

the exact KKT threshold of the fully optimized all-common subproblem. We use it only as a KKT-motivated path scale. An active cap additionally introduces a normal-cone term, and the scale is not a global threshold for the observed mixture objective. The path is

$$\Lambda^{(a)} = \{0\} \cup \left\{ \tilde{\lambda}_{\max, \eta}^{(a)} 10^{-4+4q/(L-2)} : q = 0, \dots, L-2 \right\}, \quad (16)$$

with $L = 240$. If $\tilde{\lambda}_{\max, \eta}^{(a)} > 0$, the 239 positive values range from $10^{-4}\tilde{\lambda}_{\max, \eta}^{(a)}$ to $\tilde{\lambda}_{\max, \eta}^{(a)}$ on a logarithmic grid. E-CGL and E-ACGL use the same relative range and grid size; their absolute penalty values differ because the path scale contains the fixed method-specific weights. If the endpoint is zero, the implementation sets $\Lambda^{(a)} = \{0\}$ and proceeds directly to candidate refitting. Warm starts are used in increasing penalty order.

Let

$$\mathcal{I}_{\text{record}}^{(a)} = \left\{ \ell : \hat{\Theta}_{\ell}^{(a)} \text{ is finite, nonfailed, and cap-feasible} \right\}.$$

A fit at its iteration limit is logged as unconverged but is retained in the provisional candidate family; a numerical failure is logged and skipped. Three consecutive numerical failures terminate the path. The path also stops after storing a support of size at most one, while the all-common support is always included. This is a computational heuristic, not a nested-path claim, so nonempty selection is conditional on the finite recorded path. In a targeted full-path audit reported in the Supplement, continuing six representative E-series fits through all remaining tuning values produced neither a new support nor re-entry above one coordinate.

3.3.2. Target-preserving support-constrained refit

The penalized path identifies candidate supports. For a candidate support S , the support-constrained refit omits the sparsity penalty, retains the finite concentration cap, and uses the parameterization

$$\eta_{kj} = b_j + c_{kj}, \quad \sum_{k=1}^K c_{kj} = 0, \quad c_{kj} = 0 \quad (j \notin S). \quad (17)$$

Hence $j \notin S$ implies $\eta_{1j} = \dots = \eta_{Kj} = b_j$: an inactive posterior-score contrast coordinate retains its estimated common baseline in the natural-parameter space.

The two stages are separated because the group penalty remains active on nonzero contrast groups. For any exact cap-inactive stationary point $\Theta_\ell^{(a),\circ}$ of the penalized objective at $\lambda_{\eta,\ell} > 0$, with $\|H_K E_{\ell,j}^{(a),\circ}\|_2 > 0$, the contrast-space score satisfies

$$H_K \nabla_{E,j} \ell \left(\Theta_\ell^{(a),\circ} \right) = \lambda_{\eta,\ell} w_j^{(a)} \frac{H_K E_{\ell,j}^{(a),\circ}}{\|H_K E_{\ell,j}^{(a),\circ}\|_2}. \quad (18)$$

Thus the penalized stationary point does not satisfy the active-contrast score equation of the corresponding support-constrained likelihood. This equality displays the sparsity-penalty term omitted during refitting. The two-stage construction has a Gaussian-mixture precedent in Lasso–MLE (Meynet, 2012; Meynet and Maugis-Rabusseau, 2012) and regression analogies in the relaxed Lasso and least-squares post-Lasso (Meinshausen, 2007; Belloni and Chernozhukov, 2013). These precedents motivate but do not theoretically validate the present refit. The vMF refit carries no claim of finite-sample bias reduction or risk improvement, valid post-selection inference, or global optimality.

For each active coordinate, the first $K - 1$ contrasts are free and the last is set to their negative sum. Conditional on the responsibilities, the natural-parameter update targets maximization, over $b \in \mathbb{R}^d$ and the free active contrasts, of

$$Q_S(\eta) = \sum_{k=1}^K \left[\eta_k^\top r_k + N_k \log C_d(\|\eta_k\|_2) \right] \quad (19)$$

The path fit is projected onto the support constraint and, when needed, scaled to the concentration cap. The smooth constrained M-step uses the limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm with bounds (L-BFGS-B) (Byrd et al., 1995) and analytic gradients, followed by step halving until the row-norm cap and observed-likelihood guard hold. This quasi-Newton routine is used only for refitting; the penalized E-series update remains proximal. The implementation-specific refit limits and stopping tolerances are reported under “E-series numerical controls” in the Supplementary Material.

A refit is eligible for BIC comparison only if it is finite, converged, and nonfailed; its inner optimizer reports success; it satisfies the centered-support constraint to numerical tolerance; and it has $\max_k \hat{\kappa}_k < 0.99\kappa_{\max}$. The explicit empty-support refit is initialized from the recorded fit with the smallest

support; ties use the largest observed log-likelihood and then the fixed path order.

The resulting numerically converged support-constrained refit retains the common baseline; no global maximum of the nonconvex mixture likelihood is claimed.

3.3.3. Support selection by BIC after refit

Following the BIC principle of Schwarz (1978), let $m_S = |S|$. Conditional on the path-generated candidate collection, scoring the refitted rather than penalized likelihood removes the direct effect of λ_η -dependent shrinkage from the parameter estimates used in the BIC criterion. The practical E-series nominal-dimension approximation is

$$\text{df}_E(S) = d + (K - 1)m_S + (K - 1)\mathbf{1}(m_S > 0). \quad (20)$$

The three terms count the coordinatewise common baselines, the active centered contrasts, and the mixing proportions when nonzero component contrasts are present. When $S = \emptyset$, all component densities coincide and the mixing proportions are not identifiable from the observed density, so they are not counted. On a generic identifiable stratum, this is a nominal parameter dimension. We use it as a practical within-method BIC criterion, not as an exact Schwarz marginal-likelihood approximation or an effective degrees-of-freedom result. The candidate collection is data-dependent, the constrained refits are local numerical solutions, and mixture strata can be singular at component mergers, boundary mixing weights, or the all-common model (Drton and Plummer, 2017).

For each recorded path point, define

$$\begin{aligned} S_\ell^{(a)} &= \left\{ j : \|H_K \hat{E}_{\ell,j}^{(a)}\|_2 > \varepsilon_{\text{supp}} \right\}, \\ \mathcal{S}_{\text{path}}^{(a)} &= \text{ordered unique } \left\{ S_\ell^{(a)} : \ell \in \mathcal{I}_{\text{record}}^{(a)} \right\}, \\ \mathcal{S}_{\text{cand}}^{(a)} &= \mathcal{S}_{\text{path}}^{(a)} \cup \{\emptyset\}, \\ \mathcal{S}_{\text{elig}}^{(a)} &= \left\{ S \in \mathcal{S}_{\text{cand}}^{(a)} : \hat{\Theta}_S^{\text{refit}} \text{ passes all numerical checks} \right\}. \end{aligned} \quad (21)$$

Candidate construction, refit scoring, and final selection are then

$$\begin{aligned} \text{BIC}^{\text{refit}}(S) &= -2\ell(\hat{\Theta}_S^{\text{refit}}) + \log(n) \text{df}_E(S), \\ \hat{S}^{(a)} &= \arg \min_{S \in \mathcal{S}_{\text{elig}}^{(a)}} \text{BIC}^{\text{refit}}(S). \end{aligned} \quad (22)$$

The all-common candidate is always refitted and enters the comparison if eligible. Nonempty candidates remain conditional on the finite path. For a nonempty population support, the event

$$S_\eta^\star \in \mathcal{S}_{\text{path}}^{(a)}$$

defines path coverage. Conditional BIC success evaluates exact recovery given this event, whereas overall exact recovery does not condition on it. For $S_\eta^\star = \emptyset$, coverage equals one by construction and exact recovery is $\widehat{S}^{(a)} = \emptyset$.

When several path fits generate the same support, the one with the largest observed log-likelihood is its penalized source and initializes the constrained refit. If the source of the provisional BIC winner is unconverged, the complete path is rebuilt with the extended allowance described above, unconverged sources are excluded, and refit-BIC selection is repeated. Exact BIC ties use the first eligible candidate in the fixed order. Neither rule uses class labels.

The tuning parameter is therefore treated only as a candidate-generation index. BIC after refit selects a distinct support and does not define or require a uniquely interpreted estimate of λ_η .

Each procedure has its own support target and nominal dimension; BIC selects refitted candidates within method, and cross-method values are descriptive.

EBIC (Chen and Chen, 2008) with $\gamma_{\text{EBIC}} \in \{0.25, 0.5, 1\}$ and alternative degrees-of-freedom rules are sensitivity analyses; they do not remove mixture singularity, local optimization, or data-dependent candidate construction.

Algorithm 1 writes out the E-step, the guarded proximal generalized M-step, path construction, and refit-based support selection. E-CGL is obtained with unit weights; E-ACGL uses the fixed dense-pilot weights in (1). All symbols and updates in the listing are defined in the preceding subsections. The fixed- λ subroutine returns **FAIL** only after a numerical guard or finite-value check fails. Reaching an iteration limit returns a nonfailed fit with its convergence flag set to false, so that failure and nonconvergence remain distinct; Step 3 retains the latter provisionally and Step 4 validates the selected source. Lines 4–17 of the listing define the fixed- λ_η subroutine $\text{Fit}_{\lambda_\eta}^{(a)}$ used in the subsequent path steps.

Algorithm 1. Guarded proximal GEM and support selection

Require: row-normalized $X \in \mathbb{R}^{n \times d}$; K ; $a \in \mathcal{A}$; dense starts; $L = 240$; $\kappa_{\max}$; iteration limits $T_0 = 100$, $T_{\text{retry}} = 440$, and $U_{\max}$; the stopping tolerances in Eqs. (12)–(13); ε_{maj} , ε_{acc} , $\varepsilon_{\text{supp}}$; and the backtracking limit.

Ensure: $\hat{S}^{(a)}$ and $\hat{\Theta}_{\hat{S}^{(a)}}^{\text{refit}}$.

- 1: **Step 1: Initialization.**
 - 2: **(a) Dense pilot:** select $\hat{\Theta}^D$ by the multistart continuation-and-fallback rule above.
 - 3: **(b) Fixed weights:** compute and fix $w_j^{(a)}$, $j = 1, \dots, d$, according to Eq. (1).
 - 4: **Step 2: Fixed- λ_η proximal GEM.**
 - 5: Initialize $\Theta^{(0)}$ from the supplied warm start and set **converged** = **FALSE**.
 - 6: **for** $t = 0, \dots, T_0 - 1$ **do**
 - 7: **Step 2.1: E-step.** Compute $\tau_{ik}^{(t)}$, $N_k^{(t)}$, and $r_k^{(t)}$ using Eq. (3) and the definitions immediately below it.
 - 8: **Step 2.2: Update mixing proportions.** Compute π^{cand} by Eq. (8).
 - 9: **Step 2.3: Update natural parameters.** Initialize the inner iteration at the current natural-parameter matrix.
 - 10: **for** $u = 0, \dots, U_{\max} - 1$ **do**
 - 11: $s \leftarrow (\max_k N_k^{(t)})^{-1}$, $V \leftarrow E^{(u)} - s \nabla f_t(E^{(u)})$.
 - 12: Apply the centered coordinatewise proximal update in Eq. (7) to V , for $j = 1, \dots, d$.
 - 13: Halve s and recompute V and $E^{(u+1)}$ until the majorization and concentration-cap conditions hold; if no admissible proposal is found, **return FAIL**.
 - 14: If Eq. (12) holds, **break**; **end for**; set $\Theta^+ \leftarrow (\pi^{\text{cand}}, E^{(u+1)})$.
 - 15: **Step 2.4: Acceptance and convergence.** Accept Θ^+ only if Eq. (11) holds; otherwise **return FAIL**.
 - 16: Set $\Theta^{(t+1)} \leftarrow \Theta^+$; if Eq. (13) holds, set **converged** = **TRUE** and **break**.
 - 17: **end for**; at the iteration limit, retain the current finite fit with **converged** = **FALSE**.
 - 18: **Step 3: Increasing-penalty path.**
 - 19: $\hat{\Theta}_0^{(a)} \leftarrow \text{Fit}_{\lambda_\eta=0}^{(a)}(\hat{\Theta}^D; w^{(a)})$; $\hat{\Theta}_0^{(a)} = \text{FAIL} \Rightarrow$ **return FAIL**.
 - 20: Set $\mathcal{I}_{\text{record}}^{(a)} \leftarrow \{0\}$ and form $S_0^{(a)}$ from $\hat{\Theta}_0^{(a)}$ by Eq. (21).
 - 21: Construct the KKT-motivated scale $\tilde{\lambda}_{\max, \eta}^{(a)}$ and the single path $\Lambda^{(a)}$ from Eqs. (15)–(16).
 - 22: **for** each positive $\lambda_{\eta, \ell} \in \Lambda^{(a)}$ in increasing order **do** initialize Step 2 from the preceding accepted fit.
 - 23: Repeat Step 2. If the fit is finite, nonfailed, and cap-feasible, include ℓ in $\mathcal{I}_{\text{record}}^{(a)}$ and form $S_\ell^{(a)}$ by Eq. (21).
 - 24: Log an iteration-limit fit as unconverged. On numerical failure, retain the last accepted warm start and continue; three consecutive failures terminate the path. After recording $|S_\ell^{(a)}| \leq 1$, stop the path. **end for**
 - 25: **Step 4: Target-preserving refit and support selection.**
 - 26: Construct $\mathcal{S}_{\text{cand}}^{(a)}$ by Eq. (21); for each candidate, compute $\hat{\Theta}_{S^{(a)}}^{\text{refit}}$ under Eq. (17), apply the stated numerical checks, and then form $\mathcal{S}_{\text{elig}}^{(a)}$.
 - 27: Select $\hat{S}^{(a)}$ by Eq. (22); if $\mathcal{S}_{\text{elig}}^{(a)} = \emptyset$, **return FAIL**.
 - 28: If the penalized source of $\hat{S}^{(a)}$ is unconverged, rebuild Step 3 with $T_0 + T_{\text{retry}}$, retain converged path sources only, and repeat Steps 4 and 27.
 - 29: **return** $(\hat{S}^{(a)}, \hat{\Theta}_{\hat{S}^{(a)}}^{\text{refit}})$.
-

In the proximal update, the centered term is zero when $H_K V_{\cdot j} = 0$. The guard logic is common to both methods, whereas the penalized criteria evaluated by the guards are method-specific through $w_j^{(a)}$. Refit BIC compares eligible supports only within a method.

3.3.4. Selection of the number of components

Component-number selection is not part of the E-CGL support estimator. The simulations therefore fix K at its generating value. When K is unknown, we first fit dense multistart mixtures over a prespecified set $\mathcal{K}$ and choose a resolution using a predeclared label-free density or stability criterion oriented so that smaller values are preferred. A stability score is therefore entered through its negative or another decreasing transformation. E-CGL support selection is then performed at that resolution:

$$\widehat{K} = \arg \min_{K \in \mathcal{K}} \mathcal{C}_{\text{dense}}(K), \quad \widehat{S}_{\widehat{K}} = \arg \min_{S \in \mathcal{S}_{\text{elig}}^{(\text{CGL})}(\widehat{K})} \text{BIC}^{\text{refit}}(\widehat{K}, S).$$

This two-stage diagnostic has no component-number consistency claim; competing resolutions should be reported when density fit and stability disagree.

3.3.5. Computational implementation

The implementation uses log-sum-exp stabilization in the E-step, warm starts along the penalty path, and explicit convergence and constraint checks in the centered-support refit. Optional compiled helpers use the Rcpp interface (Eddelbuettel and François, 2011) for low-level numerical operations without changing the statistical algorithm. R-only and compiled-helper outputs were checked to numerical tolerance before runtime comparisons.

The current computational claims are limited to empirical equality and runtime diagnostics. The method does not rely on an Rcpp implementation for its statistical definition.

The finite parameter space $\mathcal{T}_{\kappa_{\max}}$ is used in dense initialization, the penalized path, and support-constrained refitting. Backtracking rejects any proposed E-series iterate outside this space. A converged refit that reaches 99% of the bound is retained for diagnosis but is not eligible for information-criterion selection. This guard defines the finite computational problem; it is not an additional sparsity penalty.

3.4. Structural properties

The following statements justify the centered support estimand for a fixed number of components. They do not imply uniqueness of a fitted mixture, which remains subject to finite-mixture identifiability and label permutation. Full proofs are given in Appendix Appendix A.

Proposition 1 (Centered decomposition and pairwise support). *For any $v \in \mathbb{R}^K$, there is a unique decomposition*

$$v = \bar{v}\mathbf{1}_K + c, \quad \bar{v} = K^{-1}\mathbf{1}_K^\top v, \quad \mathbf{1}_K^\top c = 0.$$

For $v = \eta_{\cdot j}$, the following statements are equivalent:

$$c = 0; \quad \eta_{1j} = \cdots = \eta_{Kj}; \quad \eta_{kj} - \eta_{\ell j} = 0 \quad \text{for all } k < \ell.$$

Moreover,

$$\|c\|_2^2 = \frac{1}{K} \sum_{k < \ell} (\eta_{kj} - \eta_{\ell j})^2.$$

Consequently,

$$S_\eta = \bigcup_{k < \ell} \{j : \eta_{kj} - \eta_{\ell j} \neq 0\},$$

the union of the feature-dependent linear supports in all pairwise posterior log-odds.

Proposition 1 clarifies the scope of the estimand. If $c_{\cdot j} = 0$, the coefficient of x_j cancels from every pairwise posterior log-odds. The common baseline is nevertheless retained in η_k and may affect the intercept through $\|\eta_k\|_2$; it is therefore not irrelevant to the component density.

Proposition 2 (Natural-parameter and directional decomposition). *Let $\kappa = (\kappa_1, \dots, \kappa_K)^\top$, $\bar{\kappa} = K^{-1}\mathbf{1}_K^\top \kappa$, and $h = \kappa - \bar{\kappa}\mathbf{1}_K$. Write*

$$\bar{\mu} = K^{-1}M^\top \mathbf{1}_K, \quad U = H_K M.$$

Then $M = \mathbf{1}_K \bar{\mu}^\top + U$ and

$$H_K E = \bar{\kappa}U + h\bar{\mu}^\top + H_K \text{diag}(h)U.$$

The three terms represent directional heterogeneity, concentration heterogeneity along the mean directional loading, and their interaction. For support interpretations involving M and S_μ , assume $\kappa_k > 0$ for every k .

If $\kappa_1 = \cdots = \kappa_K = \kappa > 0$, then $H_K E = \kappa H_K M$ and $S_\eta = S_\mu$. If $\mu_1 = \cdots = \mu_K = \bar{\mu}$ and $h \neq 0$, then $S_\mu = \emptyset$ while $S_\eta = \text{supp}(\bar{\mu})$. Conversely, heterogeneous concentrations can make $H_K M_{\cdot j} \neq 0$ even when $H_K E_{\cdot j} = 0$. In the general decomposition the three terms may cancel, so S_η is not identified with the union of their separate supports.

Proposition 3 (Label equivariance and support invariance). *Let P be a $K \times K$ permutation matrix. Then*

$$H_K P = P H_K, \quad \|H_K P \eta_j\|_2 = \|H_K \eta_j\|_2.$$

The penalties $\mathcal{P}_a$, $a \in \mathcal{A}$, are invariant to component relabeling, their proximal maps are equivariant, and S_η is unchanged. For E-ACGL, the dense pilot is relabeled consistently before its fixed weights are computed.

For a nonzero natural parameter, the component representation is recovered as $\kappa_k = \|\eta_k\|_2$ and $\mu_k = \eta_k / \|\eta_k\|_2$. At $\eta_k = 0$, the vMF component is uniform and its direction is not identified. These algebraic statements do not establish finite-mixture identifiability or uniqueness of the fitted mixture. Component-specific interpretation assumes distinct component distributions and positive mixing weights and is made only up to label permutation. If two components have identical natural parameters, their individual mixing weights are not separately identified. Support-recovery statements are conditional on fixed K .

3.5. Algorithmic properties

Proposition 4 (Closed-form proximal update). *Let $J_K = K^{-1} \mathbf{1}_K \mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For $\xi \geq 0$ and $v \in \mathbb{R}^K$,*

$$\text{prox}_{\xi \|H_K \cdot\|_2}(v) = J_K v + \left(1 - \frac{\xi}{\|H_K v\|_2}\right)_+ H_K v,$$

where the second term is defined as zero when $H_K v = 0$.

The common baseline is unchanged. Group soft-thresholding is applied only to the centered contrast, matching the coordinate-level definition of S_η .

Proposition 5 (Smooth majorization and proximal descent). *Fix responsibilities with $N_k > 0$. The function f_t in Section 3.2.1 is convex and has a Lipschitz gradient with constant $L_t \leq \max_k N_k$. Hence, for $s \leq 1/L_t$, the proximal-gradient update in Section 3.2.2 does not increase $F_t^{(a)}$ in exact arithmetic. More generally, an update accepted with majorization tolerance ε_{maj} satisfies*

$$F_t^{(a)}(E^{(u+1)}) \leq F_t^{(a)}(E^{(u)}) + \varepsilon_{\text{maj}}.$$

The backtracking rule starts at $s = 1/\max_k N_k$ and halves the step until this majorization condition holds numerically.

Theorem 1 (Fixed-responsibility proximal convergence). *Let $d \geq 2$. Fix responsibilities such that $N_k > 0$ and $\|r_k\|_2 < N_k$ for every k , fix $a \in \mathcal{A}$ with finite positive weights, and fix a finite $\lambda_\eta \geq 0$. Then $F_t^{(a)}$ on $\mathbb{R}^{K \times d}$ is strictly convex and coercive and therefore has a unique unconstrained minimizer E_t^* . If E_t^* lies strictly inside the concentration bound, it is also the unique solution of the bounded subproblem. The unconstrained proximal-gradient method with the exact majorization backtracking rule ($\varepsilon_{\text{maj}} = 0$) converges to E_t^* , and its objective gap is $O(u^{-1})$ after u inner iterations.*

Strict convexity follows from the positive-definite vMF covariance at finite natural parameters. The condition $\|r_k\|_2 < N_k$ makes the radial linear term dominate the logarithmic Bessel term, yielding coercivity; the global Hessian bound then gives the stated proximal-gradient result. The complete argument is given in Appendix Appendix A.

The theorem concerns the limiting fixed-responsibility problem with an inactive cap. Finite inner tolerances are inexact, and an active cap would require projected-proximal analysis (Beck and Teboulle, 2009; Parikh and Boyd, 2014).

Proposition 6 (Penalized GEM monotonicity). *Let $\tau^{(t)}$ be the posterior responsibilities under $\Theta^{(t)}$. If a candidate $\Theta^{(t+1)}$ satisfies*

$$Q_{\lambda_\eta}^{(a)}(\Theta^{(t+1)} \mid \tau^{(t)}) \geq Q_{\lambda_\eta}^{(a)}(\Theta^{(t)} \mid \tau^{(t)}),$$

then, in exact arithmetic,

$$\mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t+1)}) \geq \mathcal{L}_{\lambda_\eta}^{(a)}(\Theta^{(t)}).$$

The implementation permits decreases of at most 10^{-8} , so finite traces are only numerically near-monotone. Exact monotonicity alone does not imply parameter convergence or global optimality; the next theorem instead gives a conditional stationarity result.

For a responsibility array τ , let f_τ denote the smooth fixed-responsibility objective in Section 3.2.1. Define

$$\begin{aligned}\mathcal{G}_s(E; \tau) &= \frac{1}{s} \left[E - \text{prox}_{s\lambda_\eta \mathcal{P}_a} \{E - s\nabla f_\tau(E)\} \right], \\ \mathcal{R}_\pi(\pi; \tau) &= \pi - \frac{N(\tau)}{n}, \quad N_k(\tau) = \sum_{i=1}^n \tau_{ik}.\end{aligned}\tag{23}$$

For Theorem 2, s_t is the diagnostic step size used to evaluate the natural-parameter residual at the final accepted inner iterate; it need not equal every trial step used during backtracking.

Theorem 2 (Conditional stationarity under vanishing block residuals). *Fix $a \in \mathcal{A}$ with finite positive weights and fix a finite $\lambda_\eta \geq 0$. Consider an idealized GEM sequence $\{\Theta^{(t)}\}$ for which*

$$\begin{aligned}\tau^{(t)} &= \tau(\Theta^{(t)}), \quad \Theta^{(t)}, \Theta^{(t+1)} \in \mathcal{C} \subseteq \text{ri}(\mathcal{T}_{\kappa_{\max}}), \\ \inf_{t,k} N_k(\tau^{(t)}) &> 0, \quad 0 < \underline{s} \leq s_t \leq \bar{s} < \infty, \\ \|\Theta^{(t+1)} - \Theta^{(t)}\| &\longrightarrow 0,\end{aligned}\tag{24}$$

where ri denotes relative interior, the norm on Θ is the Euclidean product norm, and $\mathcal{C} \subseteq \text{ri}(\mathcal{T}_{\kappa_{\max}})$ means that $\mathcal{C}$ is a compact subset of that relative interior. Suppose further that the block residuals satisfy

$$\|\mathcal{G}_{s_t}(E^{(t+1)}; \tau^{(t)})\|_F + \|\mathcal{R}_\pi(\pi^{(t+1)}; \tau^{(t)})\|_2 \longrightarrow 0.\tag{25}$$

Then every accumulation point $\bar{\Theta} = (\bar{\pi}, \bar{E})$ satisfies the interior KKT conditions

$$\begin{aligned}0 &\in -\nabla_E \ell(\bar{\Theta}) + \lambda_\eta \partial \mathcal{P}_a(\bar{E}), \\ \nabla_\pi \ell(\bar{\Theta}) &= \nu \mathbf{1}_K, \quad \mathbf{1}_K^\top \bar{\pi} = 1, \quad \text{for some } \nu \in \mathbb{R}.\end{aligned}\tag{26}$$

Thus every accumulation point is stationary for the finite-parameter penalized observed objective. The stationarity statement is equivariant under component-label permutation.

The proof passes to a convergent subsequence, uses continuity of the responsibility and proximal maps, and applies Fisher’s identity to obtain the observed-score KKT system. Details are given in Appendix Appendix A.

This conditional local result is not a convergence claim for every finite-tolerance path. Active bounds, failed path points, nonvanishing residuals, or component degeneracy fall outside its assumptions; see (Dempster et al., 1977; Green, 1990; Wu, 1983; Tseng, 2004) for related EM convergence theory.

3.6. Computational complexity

For dense X , one E-step costs $O(nKd)$; for a sparse matrix it costs $O\{K \text{nnz}(X)\}$, where $\text{nnz}(X)$ is the number of nonzero entries of X . Forming the sufficient statistics has the same order, and one centered proximal-gradient update costs $O(Kd)$. Excluding the input matrix, the main working memory is $O(nK + Kd)$, in addition to $O(nd)$ for dense X or $O\{\text{nnz}(X)\}$ for sparse X . If the single path evaluates L tuning points, let T_{GEM} denote the average number of outer GEM iterations per path point and T_{PG} the average number of inner proximal-gradient updates per GEM iteration. Its leading cost is

$$O[LT_{\text{GEM}}\{K \text{nnz}(X) + T_{\text{PG}}Kd\}]$$

for sparse input, plus the costs of distinct-support refits. These are operation counts rather than runtime superiority claims. Both E-CGL and E-ACGL use $L = 240$ before early stopping. Multiple dense starts add their own E-step and M-step costs before the path is constructed.

4. Numerical studies

4.1. Design and evaluation criteria

The primary experiment evaluates recovery of posterior-score contrast support under varying overlap and concentration heterogeneity. Observations are drawn from a balanced $K = 4$ vMF mixture in $d = 200$ dimensions. The coordinates are partitioned as

$$\{1, \dots, d\} = S_C \cup S_D \cup S_N, \quad (q_C, q_D, q_N) = (4, 16, 180),$$

where S_C contains common natural-parameter coefficients, S_D contains centered component contrasts, and S_N contains globally null coordinates. Let $q_{\text{block}} = 4$ be the number of contrast coordinates assigned to each component block. For a contrast coordinate j in block g , define

$$u_{kj} = \begin{cases} 1, & k = g, \\ -1/(K - 1), & k \neq g. \end{cases}$$

and set $u_k = (u_{kj} : j \in S_D) \in \mathbb{R}^{q_D}$. For a candidate common-background norm satisfying $0 \leq A < \min_{1 \leq k \leq K} \kappa_k$, the natural parameters are generated

by

$$\eta_{kj}(A) = \begin{cases} A/\sqrt{q_C}, & j \in S_C, \\ \alpha_k(A)u_{kj}, & j \in S_D, \\ 0, & j \in S_N, \end{cases} \quad \alpha_k(A) = \frac{\sqrt{\kappa_k^2 - A^2}}{\|u_k\|_2}.$$

Hence $\|\eta_k(A)\|_2 = \kappa_k$ and $\mu_k(A) = \eta_k(A)/\kappa_k$. The clustering difficulty is calibrated by the oracle Bayes error

$$e_B = \Pr_{\Theta^*} \left\{ \arg \max_k \Pr(Z = k \mid X; \Theta^*) \neq Z \right\}.$$

We use $e_B \in \{0.025, 0.05, 0.10\}$, $n \in \{300, 1000\}$, and either

$$\kappa = (45, 45, 45, 45) \quad \text{or} \quad \kappa = (30, 40, 50, 60).$$

The common-to-contrast allocation is calibrated by Monte Carlo bisection while the prescribed concentrations and support sets are held fixed. Independent validation samples of size 200,000 gave achieved errors between 0.02566 and 0.10035; the largest absolute discrepancy from its target was 0.00172.

Each of the 12 cells contains 100 paired repetitions and an independent test sample of size 5,000. The common comparison set consists of spherical k -means, dense vMF mixtures with shared and component-specific concentrations, the Rossi-style entrywise mean-direction lasso (M-L), E-CGL, and E-ACGL. The centered mean-direction group lasso (M-CGL) is included as a directional companion in the four prespecified $e_B = 0.05$ cells. E-CGL and E-ACGL use the single 240-point, zero-augmented KKT-motivated path in (16). M-CGL uses a separate 240-point dense-score-proxy path from $10^{-4}\lambda_{\text{proxy},\mu}$ to $1.25\lambda_{\text{proxy},\mu}$, where the proxy scale is defined in Supplementary Section S8. M-L retains its Rossi-style event-driven path, with at most 240 path points, including the zero-penalty fit, in the simulation, because its threshold events are not indexed by either of the centered-group path scales. All mixture fits use 10 dense starts and Rcpp helpers; sparse mixture fits use BIC after target-specific support refitting. Across the common 24-cell rerun—the 12 primary cells, three additional estimand-separation cells, four $e_B = 0.05$ sample-size cells, and five stress cells—four of the 2,400 dataset-replicate dense free-concentration pilots shared by the two E-series methods reached the 200-iteration limit. Continuing each terminal iterate to convergence left all eight corresponding E-CGL and E-ACGL supports unchanged;

Table 1: Methods, support targets, and roles in the numerical comparisons.

Method	Support target	Role
Spherical k -means	none	external clustering baseline
Dense vMF, shared κ	none	concentration-restricted baseline
Dense vMF, free κ_k	none	likelihood baseline
M-L	S_P	published prototype-sparsity reference
M-CGL	S_μ	directional companion
E-CGL	S_η	primary proposed estimator
E-ACGL	S_η	secondary adaptive extension

the largest absolute BIC change was 0.027. An exhaustive audit of the selected penalized source in all 4,800 E-series method–replicate fits identified 37 sources that had reached the original 100-iteration limit; 32 selected supports had no converged source on that recorded path. We therefore reran only those 37 method–replicate paths with a total limit of 540 outer iterations per point and admitted converged path points only. All 37 corrected sources and refits converged; 34 supports changed, including 32 changes in support size. The primary 12-cell results were unaffected. All additional and stress results reported below use the corrected archive; full audit details are given in the Supplementary Material.

Table 1 summarizes the estimand and role of each method. The common comparison set is evaluated in all 12 primary cells; M-CGL is restricted to the four prespecified $e_B = 0.05$ companion cells.

For clarity, the two mean-direction comparators use different penalty functionals. The Rossi-style entrywise prototype penalty is

$$\mathcal{P}_{\text{M-L}}(M) = \sum_{k=1}^K \sum_{j=1}^d |\mu_{kj}|, \quad \|\mu_k\|_2 = 1,$$

whereas the directional companion uses

$$\mathcal{P}_{\text{M-CGL}}(M) = \sum_{j=1}^d \|c_{\cdot j}^{(\mu)}\|_2, \quad \|\mu_k\|_2 = 1.$$

The respective fitted criteria subtract $\lambda_\mu \mathcal{P}_{\text{M-L}}(M)$ and $\lambda_\mu \mathcal{P}_{\text{M-CGL}}(M)$ using their method-specific paths.

In the primary simulations, M-L estimates component-specific concentrations so that prototype selection is not coupled to a false shared-concentration

restriction. The Rossi bridge and real-data M-L fit instead use the published shared-concentration specification; their results are conditional on that restriction. M-CGL introduces an auxiliary matrix $\Xi \in \mathbb{R}^{K \times d}$ subject to $\Xi = H_K M$, thereby separating a product-of-spheres direction update from centered group thresholding and a scaled-dual update before numerical concentration updates. At feasibility, $\Xi_{\cdot j} = c_{\cdot j}^{(\mu)}$. Its target-preserving refit enforces $c_{\cdot j}^{(\mu)} = 0$ outside the selected directional-contrast support. The augmented-Lagrangian updates and numerical residual criteria are given in Supplementary Section S8, *Directional companion implementation*.

Clustering agreement is evaluated by the adjusted Rand index (ARI) (Hubert and Arabie, 1985) and normalized mutual information (NMI) (Strehl and Ghosh, 2002). The implementation uses geometric-mean entropy normalization for NMI; larger values indicate closer agreement. Predictive fit is evaluated by the per-observation test negative log-likelihood (NLL)

$$\text{NLL}_{\text{test}} = -\frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \log f(X_i; \hat{\Theta}),$$

where f is the vMF mixture density with respect to surface-area measure on the unit sphere. Such a density can exceed one, so its reported NLL can be negative. Let $\hat{S}$ denote the coordinate set selected by the method under evaluation. For the E-series methods, $\hat{S} = \hat{S}_\eta$; for M-L and M-CGL, it estimates S_P and S_μ , respectively. Posterior-score contrast support recovery is summarized against $S_\eta^* = S_D$, first by the selected-coordinate counts

$$\hat{q}_C = |\hat{S} \cap S_C|, \quad \hat{q}_D = |\hat{S} \cap S_D|, \quad \hat{q}_N = |\hat{S} \cap S_N|,$$

and then by the true-positive rate (TPR), false-positive rate (FPR), precision, and F_1 score, together with the exact-support rate. Using $(\eta^c)_{kj} = c_{kj}$ for a centered natural-parameter entry, write $c_{kj}^* = \eta_{kj}^* - K^{-1} \sum_h \eta_{hj}^*$ and $\hat{c}_{kj} = \hat{\eta}_{kj} - K^{-1} \sum_h \hat{\eta}_{hj}$. Centered natural-parameter estimation is summarized, after MSE-optimal component alignment, by

$$\text{MSE}_{\eta^c} = \min_{P \in \mathfrak{P}_K} \frac{1}{Kd} \|PH_K \hat{E} - H_K E^*\|_F^2,$$

where $\mathfrak{P}_K$ is the set of $K \times K$ permutation matrices. Selected table entries are accompanied by Monte Carlo standard errors (MCSEs), computed across the 100 repetitions in each cell. The primary support tables evaluate every sparse

procedure against S_η^* to answer a common posterior-score contrast question. Those entries are therefore cross-target diagnostics for M-L and M-CGL. Target-specific recovery is reported separately: M-L is evaluated against S_P in the summary below, and M-CGL is evaluated against S_μ in Section 4.3. Consequently, $F_{1,P}$, $F_{1,\mu}$, and $F_{1,\eta}$ are not interchangeable method rankings.

4.2. Posterior-score contrast support recovery

Tables 2 and 3 report estimation, posterior-score selection, and clustering over the 12-cell grid. Figs. 1–2 show the replication-level ARI and $\log_{10}(\text{MSE}_{\eta^c})$ distributions. Common-target support metrics use S_η^* ; M-L and M-CGL are also assessed against their own targets below.

Across 11 of the 12 primary cells, E-CGL selected between 16.02 and 16.67 coordinates and attained $F_{1,\eta} \geq 0.981$. The exception combined $n = 300$, $e_B = 0.10$, and heterogeneous concentrations. In that cell E-CGL selected 20.97 coordinates and attained $F_{1,\eta} = 0.816$; all six $n = 1000$ cells had $F_{1,\eta} \geq 0.988$. M-L retained 199.42–199.88 of 200 coordinates. Its low cross-target $F_{1,\eta}$ partly reflects a different estimand, but its own-target $F_{1,P} = 0.182$ also shows near-dense prototype selection. ARI decreased with e_B even where support recovery remained accurate. Under the common single-path rule, E-ACGL selected 15.23–16.33 coordinates at $n = 300$ and 16.01–16.06 at $n = 1000$; its corresponding $F_{1,\eta}$ ranges were 0.947–0.996 and 0.998–1.000.

4.3. Directional versus posterior-score estimands

For the simulation truths in this subsection, let $q_\mu^* = |S_\mu|$ and $q_\eta^* = |S_\eta|$.

Four diagnostics separate directional and posterior-score heterogeneity. Under common concentration, $S_\mu = S_\eta$. Under pure concentration heterogeneity, the component directions are common, so $S_\mu = \emptyset$ while S_η contains 16 coordinates; this design uses $\kappa = (10, 30, 80, 200)$. The shared-natural-parameter design uses $\kappa = (30, 40, 50, 60)$ and contains 80 common natural-parameter coordinates and 20 posterior-score contrast coordinates. The crossed design uses the same heterogeneous concentrations and contains four η -only, four μ -only, eight jointly active, and eight null coordinates. Table 4 compares recovery against both population targets in these four designs.

For the empty directional target, support F_1 is undefined and “Exact support” is the probability of selecting the empty support. M-CGL selected that support in 20% of the pure-concentration repetitions, whereas E-CGL recovered the nonempty posterior-score contrast support with mean $F_{1,\eta} = 0.992$. In the crossed design, each method had high target-specific F_1 and lower

Table 2: Primary method comparison at $n = 300$. Estimation, posterior-score selection, and clustering performance are reported over 100 paired repetitions.

e_B	κ pattern	method	Estimation	Posterior-score selection				Clustering and fit	
			MSE $_{\eta^c}$ (100 $\times$ MCSE)	$\hat{q}$	FPR $_{\eta}$	TPR $_{\eta}$	$F_{1,\eta}$	ARI	test NLL
0.025	equal	Sph. k -means	—	—	—	—	—	0.863	—
		Dense-S	2.471 (1.652)	—	—	—	—	0.869	-245.921
		Dense-F	2.505 (1.675)	—	—	—	—	0.864	-245.906
		M-L	2.447 (1.675)	199.58	0.998	1.000	0.148	0.865	-245.934
		E-CGL	0.198 (0.402)	16.14	0.001	1.000	0.996	0.927	-246.935
		E-ACGL	0.198 (0.402)	16.14	0.001	1.000	0.996	0.927	-246.935
0.025	heterogeneous	Sph. k -means	—	—	—	—	—	0.775	—
		Dense-S	3.092 (1.986)	—	—	—	—	0.798	-245.869
		Dense-F	2.540 (1.702)	—	—	—	—	0.856	-246.110
		M-L	2.475 (1.648)	199.61	0.998	1.000	0.148	0.860	-246.138
		E-CGL	0.192 (0.512)	16.15	0.001	1.000	0.996	0.925	-247.153
		E-ACGL	0.192 (0.512)	16.15	0.001	1.000	0.996	0.925	-247.153
0.050	equal	Sph. k -means	—	—	—	—	—	0.620	—
		Dense-S	2.810 (2.275)	—	—	—	—	0.738	-245.936
		Dense-F	2.863 (2.347)	—	—	—	—	0.726	-245.919
		M-L	2.771 (2.133)	199.65	0.998	1.000	0.148	0.731	-245.954
		M-CGL	0.222 (0.585)	16.24	0.001	1.000	0.993	0.856	-247.009
		E-CGL	0.220 (0.599)	16.25	0.001	1.000	0.992	0.856	-247.010
0.050	heterogeneous	E-ACGL	0.219 (0.577)	16.23	0.001	1.000	0.993	0.856	-247.011
		Sph. k -means	—	—	—	—	—	0.580	—
		Dense-S	3.764 (4.549)	—	—	—	—	0.649	-245.860
		Dense-F	3.077 (4.439)	—	—	—	—	0.709	-246.082
		M-L	2.957 (4.432)	199.70	0.998	1.000	0.148	0.718	-246.122
		M-CGL	0.615 (2.590)	20.90	0.034	0.917	0.797	0.815	-247.064
0.100	equal	E-CGL	0.232 (0.822)	16.67	0.004	1.000	0.981	0.857	-247.205
		E-ACGL	0.209 (0.555)	16.18	0.001	1.000	0.995	0.858	-247.215
		Sph. k -means	—	—	—	—	—	0.211	—
		Dense-S	4.191 (9.094)	—	—	—	—	0.436	-245.788
		Dense-F	4.509 (10.058)	—	—	—	—	0.390	-245.737
		M-L	4.143 (9.850)	199.67	0.998	1.000	0.148	0.422	-245.830
0.100	heterogeneous	E-CGL	0.253 (0.660)	16.28	0.002	1.000	0.992	0.724	-247.130
		E-ACGL	0.273 (1.544)	16.33	0.002	0.997	0.987	0.721	-247.124
		Sph. k -means	—	—	—	—	—	0.444	—
		Dense-S	4.768 (4.502)	—	—	—	—	0.484	-245.998
		Dense-F	4.591 (7.985)	—	—	—	—	0.512	-246.148
		M-L	4.529 (8.296)	199.62	0.998	1.000	0.148	0.514	-246.173
		E-CGL	0.871 (3.751)	20.97	0.033	0.926	0.816	0.636	-247.164
		E-ACGL	0.419 (2.072)	15.23	0.002	0.925	0.947	0.702	-247.283

Notes: Point estimates are reported on their original scales; parenthesized values are Monte Carlo standard errors of mean MSE $_{\eta^c}$ multiplied by 100 and reported to three decimal places. They are not standard errors of fitted parameters. Selection rates are on the 0–1 scale and use S_{η} for all sparse methods. Target-specific M-CGL results are reported separately. Dashes indicate that a method does not return the corresponding quantity. M-CGL was prespecified only for $e_B = 0.05$.

Table 3: Primary method comparison at $n = 1000$. Estimation, posterior-score selection, and clustering performance are reported over 100 paired repetitions.

e_B	κ pattern	method	Estimation	Posterior-score selection				Clustering and fit	
			MSE $_{\eta^c}$ (100 $\times$ MCSE)	$\hat{q}$	FPR $_{\eta}$	TPR $_{\eta}$	$F_{1,\eta}$	ARI	test NLL
0.025	equal	Sph. k -means	—	—	—	—	—	0.917	—
		Dense-S	0.676 (0.347)	—	—	—	—	0.918	-246.966
		Dense-F	0.679 (0.342)	—	—	—	—	0.917	-246.964
		M-L	0.672 (0.339)	199.42	0.997	1.000	0.149	0.918	-246.968
		E-CGL	0.055 (0.110)	16.02	0.000	1.000	0.999	0.933	-247.245
		E-ACGL	0.055 (0.110)	16.02	0.000	1.000	0.999	0.933	-247.245
0.025	heterogeneous	Sph. k -means	—	—	—	—	—	0.866	—
		Dense-S	1.147 (0.534)	—	—	—	—	0.861	-246.931
		Dense-F	0.685 (0.404)	—	—	—	—	0.912	-247.167
		M-L	0.677 (0.405)	199.57	0.998	1.000	0.148	0.913	-247.170
		E-CGL	0.056 (0.115)	16.02	0.000	1.000	0.999	0.928	-247.451
		E-ACGL	0.056 (0.115)	16.02	0.000	1.000	0.999	0.928	-247.451
0.050	equal	Sph. k -means	—	—	—	—	—	0.835	—
		Dense-S	0.731 (0.436)	—	—	—	—	0.837	-247.035
		Dense-F	0.737 (0.437)	—	—	—	—	0.836	-247.032
		M-L	0.726 (0.437)	199.58	0.998	1.000	0.148	0.837	-247.036
		M-CGL	0.061 (0.150)	16.04	0.000	1.000	0.999	0.867	-247.316
		E-CGL	0.061 (0.161)	16.06	0.000	1.000	0.998	0.867	-247.316
0.050	heterogeneous	Sph. k -means	—	—	—	—	—	0.773	—
		Dense-S	1.273 (0.642)	—	—	—	—	0.765	-247.004
		Dense-F	0.747 (0.476)	—	—	—	—	0.835	-247.228
		M-L	0.736 (0.478)	199.67	0.998	1.000	0.148	0.836	-247.232
		M-CGL	0.087 (0.143)	20.04	0.022	1.000	0.888	0.867	-247.510
		E-CGL	0.061 (0.132)	16.02	0.000	1.000	0.999	0.869	-247.517
0.100	equal	Sph. k -means	—	—	—	—	—	0.676	—
		Dense-S	0.858 (0.482)	—	—	—	—	0.684	-247.146
		Dense-F	0.866 (0.490)	—	—	—	—	0.682	-247.144
		M-L	0.851 (0.492)	199.64	0.998	1.000	0.148	0.683	-247.149
		E-CGL	0.067 (0.133)	16.02	0.000	1.000	0.999	0.747	-247.445
		E-ACGL	0.067 (0.133)	16.02	0.000	1.000	0.999	0.747	-247.445
0.100	heterogeneous	Sph. k -means	—	—	—	—	—	0.589	—
		Dense-S	1.758 (1.744)	—	—	—	—	0.595	-247.110
		Dense-F	1.013 (1.822)	—	—	—	—	0.669	-247.308
		M-L	0.970 (1.583)	199.88	0.999	1.000	0.148	0.675	-247.316
		E-CGL	0.077 (0.198)	16.41	0.002	1.000	0.988	0.753	-247.627
		E-ACGL	0.071 (0.155)	16.01	0.000	1.000	1.000	0.753	-247.630

Notes: Point estimates are reported on their original scales; parenthesized values are Monte Carlo standard errors of mean MSE $_{\eta^c}$ multiplied by 100 and reported to three decimal places. They are not standard errors of fitted parameters. Selection rates are on the 0–1 scale and use S_{η} for all sparse methods. Target-specific M-CGL results are reported separately. Dashes indicate that a method does not return the corresponding quantity. M-CGL was prespecified only for $e_B = 0.05$.

Table 4: Directional and posterior-score contrast recovery in the estimand-separation designs. Entries are means over 100 replications, and the two target-specific F_1 scores are evaluated against their respective population targets. The exact-support column uses S_μ for M-CGL and S_η for E-CGL.

Design	Method	q_μ^*	q_η^*	$\hat{q}$	$F_{1,\mu}$	$F_{1,\eta}$	Exact support	ARI
common κ	M-CGL	16	16	16.04	0.999	0.999	0.96	0.867
common κ	E-CGL	16	16	16.06	0.998	0.998	0.95	0.867
pure concentration	M-CGL	0	16	0.96	–	0.005	0.20	0.675
pure concentration	E-CGL	0	16	16.27	–	0.992	0.83	0.633
shared-natural-parameter	M-CGL	100	20	21.13	0.348	0.974	0.00	0.859
shared-natural-parameter	E-CGL	100	20	20.02	0.333	1.000	0.98	0.870
crossed support	M-CGL	12	12	11.68	0.984	0.677	0.64	0.996
crossed support	E-CGL	12	12	11.89	0.681	0.980	0.69	0.996

F_1 against the other target. The shared-natural-parameter design exposed a boundary for the directional companion: its exact support was not selected in any replication. These results motivate target-specific, rather than winner-take-all, comparisons.

4.4. Sample-size recovery and oracle benchmarks

Figure 3 traces E-CGL as the sample size increases from 300 to 2000. Centered- η MSE decreased along every trajectory. At $e_B = 0.10$ with heterogeneous concentrations, the most difficult small-sample sequence improved from $F_{1,\eta} = 0.816$ and exact recovery 0.02 at $n = 300$ to 0.999 and 0.97 at $n = 2000$; the other trajectories approached exact recovery at least as quickly.

The path oracle is the fitted candidate with the largest $F_{1,\eta}$ within each repetition. The oracle- S_η^* estimator refits under the known population support. We report

$$\Delta_{\text{sel}} = F_{1,\eta}^{\text{path oracle}} - F_{1,\eta}^{\text{BIC}}, \quad \Delta_{\text{NLL}} = \text{NLL}(\hat{\Theta}) - \text{NLL}(\hat{\Theta}_{S_\eta^*}^{\text{oracle support}}).$$

These are finite-sample benchmarks, not asymptotic oracle claims. Selector gaps were at most 0.0083 at $e_B = 0.05$. The difficult heterogeneous- κ , $e_B = 0.10$ sequence had $\Delta_{\text{sel}} = 0.0478$ and $\Delta_{\text{NLL}} = 0.1653$ at $n = 300$, with the selector gap falling to 0.0009 by $n = 2000$. At $n = 300$, the separate path-oracle gap was 0.1357: the exact support occurred on 6% of fitted paths, and BIC selected it in 33.3% of those covered repetitions. Thus candidate-path coverage, rather than BIC selection alone, was the main small-sample limitation in this cell.

4.5. Stress conditions and computation

The five stress cells cover moderately and strongly dense support, $d > n$, and weak beta-min. E-CGL attained $F_{1,\eta} = 0.998$ in the sparse weak-signal cell but 0.803 for $d = 500, n = 300$. With 80 active coordinates, $F_{1,\eta}$ improved from 0.805 at $n = 300$ to 0.901 at $n = 1000$. E-ACGL reduced false positives in some cells, with mixed effects on estimation and clustering; all results appear in Table 5.

In a paired runtime diagnostic for the representative heterogeneous $e_B = 0.05$, $n = 1000$, $d = 200$ cell, median end-to-end wall-clock times and interquartile ranges (IQRs) over 100 repetitions were 509.61 (415.62–597.18) seconds for M-CGL and 74.62 (69.14–79.85) seconds for E-CGL. The sparse-method totals include the paired dense initialization and the method-specific path, refit, and selector; one-time Rcpp compilation was excluded. Because the methods use different path and optimization rules, these values describe computational burden rather than speed superiority.

Table 5: Stress-condition results over 100 repetitions. Parenthesized scenario triplets give (n, d, q^*) .

Scenario	method	$\hat{q}$	TPR_η	FPR_η	$F_{1,\eta}$	exact	ARI	MSE_{η^c}
Moderately dense (300, 200, 80)	Sph. k -means	–	–	–	–	–	0.449	–
	Dense-S	–	–	–	–	–	0.481	4.830
	Dense-F	–	–	–	–	–	0.512	4.393
	M-L	199.50	1.000	0.996	0.572	0.00	0.515	4.360
	E-CGL	81.99	0.814	0.141	0.805	0.00	0.556	4.985
	E-ACGL	58.15	0.689	0.025	0.797	0.00	0.545	3.364
Moderately dense (1000, 200, 80)	Sph. k -means	–	–	–	–	–	0.586	–
	Dense-S	–	–	–	–	–	0.600	1.719
	Dense-F	–	–	–	–	–	0.670	0.989
	M-L	199.81	1.000	0.998	0.572	0.00	0.673	0.957
	E-CGL	80.31	0.902	0.068	0.901	0.00	0.649	0.820
	E-ACGL	67.53	0.843	0.001	0.913	0.00	0.679	0.642
Strongly dense (1000, 200, 160)	Sph. k -means	–	–	–	–	–	0.589	–
	Dense-S	–	–	–	–	–	0.598	1.716
	Dense-F	–	–	–	–	–	0.669	0.998
	M-L	199.73	1.000	0.993	0.890	0.00	0.671	0.968
	E-CGL	152.49	0.902	0.205	0.923	0.00	0.632	2.005
	E-ACGL	125.60	0.783	0.007	0.877	0.00	0.613	1.611
High-dimensional (300, 500, 40)	Sph. k -means	–	–	–	–	–	0.285	–
	Dense-S	–	–	–	–	–	0.405	9.863
	Dense-F	–	–	–	–	–	0.424	9.519
	M-L	499.99	1.000	1.000	0.148	0.00	0.431	9.441
	E-CGL	54.70	0.927	0.038	0.803	0.00	0.759	1.548
	E-ACGL	36.74	0.853	0.006	0.888	0.00	0.766	1.252
Weak beta-min (1000, 200, 16)	Sph. k -means	–	–	–	–	–	0.776	–
	Dense-S	–	–	–	–	–	0.768	1.269
	Dense-F	–	–	–	–	–	0.837	0.750
	M-L	199.72	1.000	0.998	0.148	0.00	0.838	0.740
	E-CGL	15.99	0.998	0.000	0.998	0.93	0.869	0.059
	E-ACGL	15.99	0.998	0.000	0.998	0.93	0.869	0.059

Notes: Support metrics use S_η . Dashes indicate that a method does not return a coordinate support. The strongly dense cell is a negative control rather than a setting tailored to sparse selection. Monte Carlo standard errors, test NLL, convergence, endpoint, and runtime summaries are reported in the Supplement.

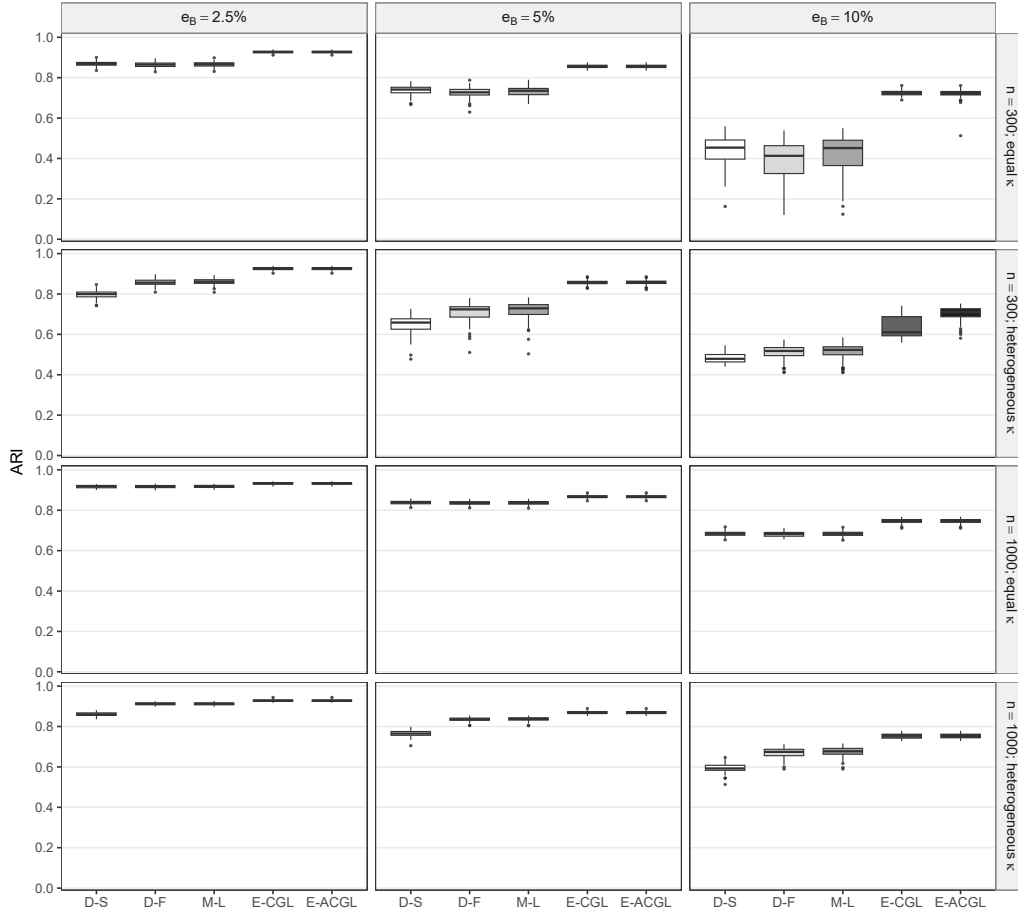


Figure 1: Clustering accuracy across the primary simulation grid. Boxplots show ARI over 100 paired replications in each cell. Columns give the target oracle Bayes error, and rows give sample size and concentration pattern. D-S and D-F denote dense vMF mixtures with shared and component-specific concentrations, respectively. M-CGL is omitted because it was evaluated only in the four prespecified $e_B = 0.05$ cells; its target-specific results are reported in Table 4. Spherical k -means is omitted from the figure for readability and is reported in Tables 2 and 3.

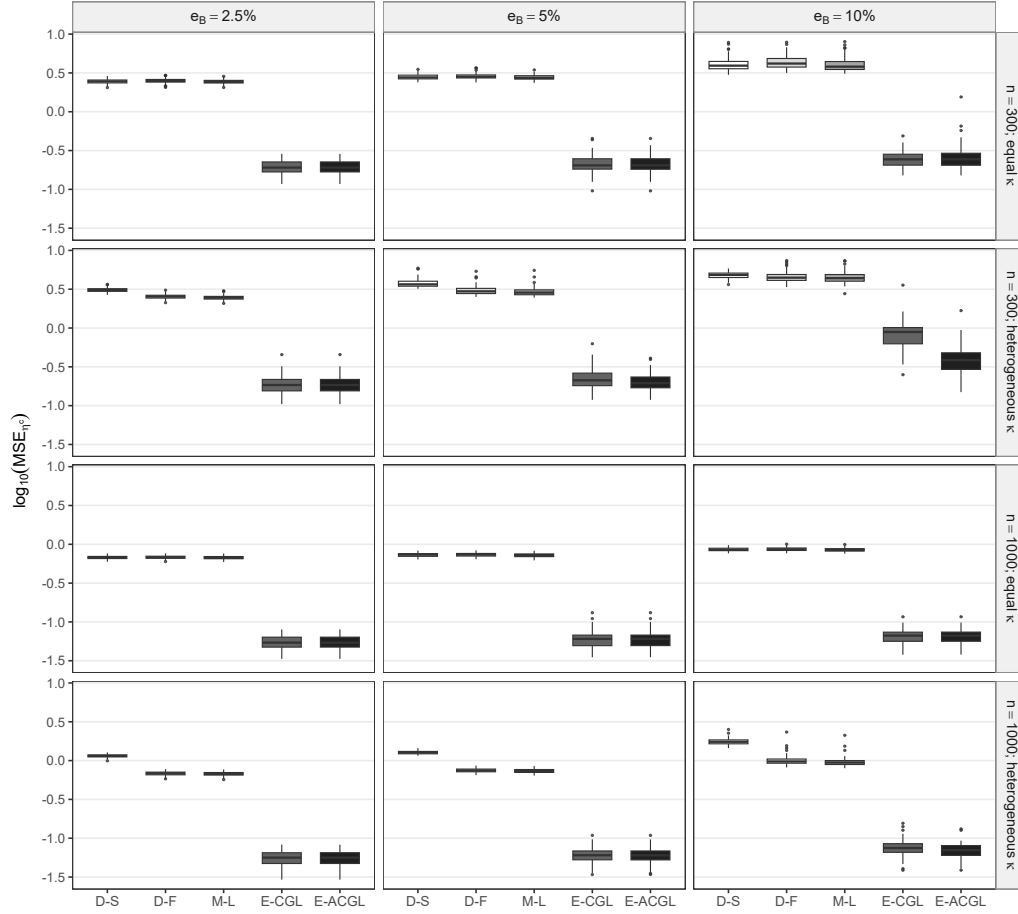


Figure 2: Centered natural-parameter estimation error across the primary simulation grid. Boxplots show $\log_{10}(\text{MSE}_{\eta^c})$ over 100 paired replications in each cell, where $\eta^c = H_K E$. Columns give the target oracle Bayes error, and rows give sample size and concentration pattern. D-S and D-F denote dense vMF mixtures with shared and component-specific concentrations, respectively. M-CGL is omitted because it was evaluated only in the four prespecified $e_B = 0.05$ cells; its target-specific results are reported in Table 4.

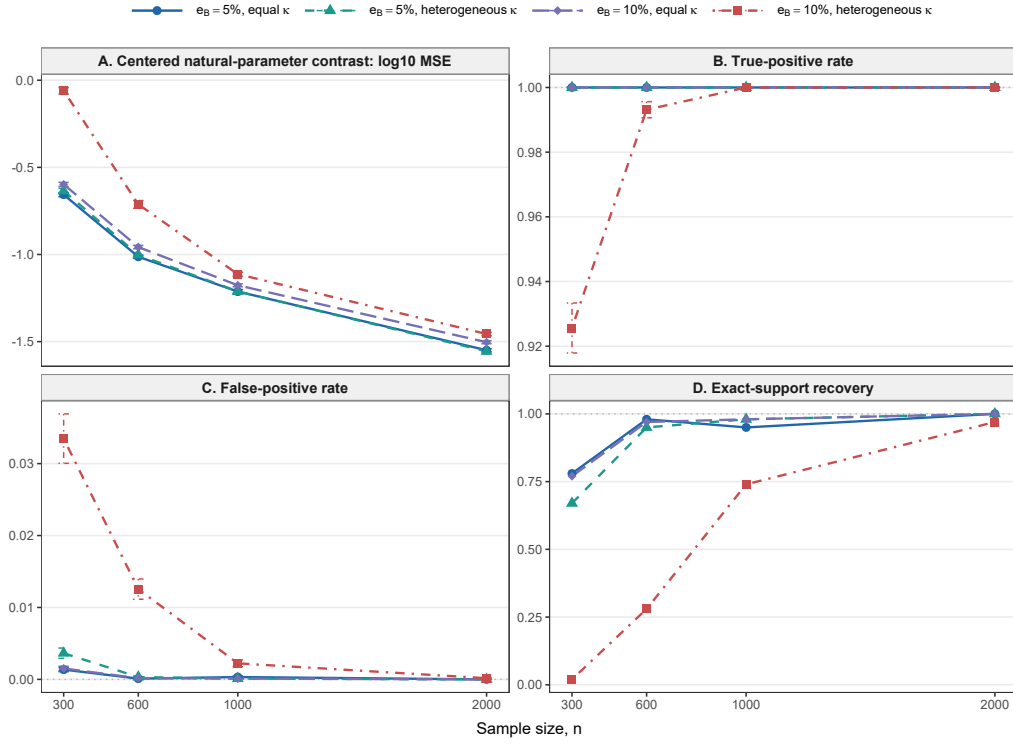


Figure 3: Sample-size trajectories for E-CGL. Points show Monte Carlo means, and error bars show Monte Carlo standard errors over 100 replications. Both oracle-error levels are shown under equal and heterogeneous concentrations.

5. Real data analysis

5.1. Data and analysis protocol

Classic3 contains abstracts from the CISI, CRAN, and MED document collections and has been used in directional document clustering (Banerjee et al., 2005). The raw payload contained $n_0 = 3,890$ documents. We removed two exact duplicates and five prespecified near-duplicates, leaving $n = 3,883$ documents. The cleaned class counts were 1,454, 1,397, and 1,032, respectively.

Documents were encoded with the vocabulary-aligned Sparse Lexical and Expansion Model (SPLADE), with a maximum input sequence length of 512 (Formal et al., 2021, 2022). The exact encoding identifiers were

Model: `naver/splade-cocondenser-ensembledistil`
Revision: `49cf4c7b0db5b870a401ddf5e2669993ef3699c7`.

The payload and vocabulary checksums are reported in Supplementary Section S9. Coordinates occurring in fewer than two documents were removed, the remaining coordinates were ranked by variance over the unlabeled cleaned payload, the top 2,000 were retained, and each vector was normalized to unit Euclidean norm. The resulting matrix had no zero rows, duplicate transformed rows, duplicate vocabulary entries, or nonfinite values. The model identifier, revision, vocabulary hash, duplicate audit, and payload hash were fixed before model fitting.

The preprocessing rules and split indices were prespecified. E-CGL uses the same single 240-point path and explicit empty-support candidate as in the primary simulations.

The benchmark resolution was fixed at $K = 3$ to match the three supplied broad topic labels. The labels were otherwise withheld from coordinate ranking, initialization, parameter estimation, tuning, and support selection; only after fitting were they used to calculate ARI and NMI and to name the aligned components. Five prespecified 80/20 splits provide stability and held-out diagnostics; they are not pooled with the full-data estimates. After the prespecified duplicate exclusions, each split ranks coordinates using its training documents only. The resulting training-selected coordinate mask is then applied without change to the test documents, after which both matrices are row-normalized.

5.2. Methods and evaluation

The comparison comprised spherical k -means, permutation-tuned sparse k -means (Witten and Tibshirani, 2010), dense vMF mixtures with shared and component-specific concentrations, the Rossi-style M-L prototype selector, M-CGL, E-CGL, and the secondary E-ACGL extension. Likelihood-based fits used 30 dense random starts. The M-L path allowed at most 600 path points, including the zero-penalty fit. E-CGL and E-ACGL used the single zero-augmented 240-point relative path defined in Section 3; M-CGL used the same 240-point dense-score-proxy path as in the numerical study. Each sparse vMF method was followed by target-specific distinct-support refitting and within-method BIC selection. M-CGL and the E-series methods also included an explicit empty-support candidate; an empty prototype support is incompatible with the unit-norm M-L directions. E-ACGL used the fixed weights in (1). The same E-series rule was used in each prespecified Classic3 split. The sparse vMF methods were selected by their method-specific BIC after support-constrained refitting. Sparse k -means used a label-blind permutation gap criterion with 25 permutations and 10 weight-bound candidates.

Selected q is target-specific: weighted coordinates for sparse k -means, prototype-union coordinates for M-L, directional-contrast coordinates for M-CGL, and posterior-score contrast coordinates for E-CGL and E-ACGL. Accordingly, differences in q do not by themselves rank the estimands. All methods were evaluated by ARI and NMI. Observed log-likelihood and the practical nominal-dimension BIC were additionally reported for vMF models.

5.3. Full-data results and interpretation

The dense free- κ_k mixture attained ARI 0.9760 and NMI 0.9538. E-CGL selected 1,420 of 2,000 coordinates, excluding 29.00% of the coordinates from posterior-score contrasts while attaining ARI 0.9761 and NMI 0.9539. M-CGL produced a closely matching partition (ARI 0.976 and NMI 0.954) with 1,471 directional-contrast coordinates. M-L retained 1,942 prototype-support coordinates. E-ACGL selected 1,438 posterior-score contrast coordinates and attained ARI 0.9746 and NMI 0.9515. The permutation-tuned sparse k -means fit selected 985 coordinates but had ARI 0.1374; its selected weight bound was not at the evaluated grid boundary. Independent restarts at that bound reproduced ARI 0.1373 and nearly the same objective value, so this result is reported as performance of the permutation-selected solution under the SPLADE representation rather than as a single-start anomaly or

a general statement about sparse k -means. Table 6 summarizes the full-data comparison.

Table 6: Clustering and selected support in the full-data Classic3 analysis. The data contain $n = 3,883$ documents and $d = 2,000$ coordinates after duplicate removal; $K = 3$, and component order is aligned to CISI, CRAN, and MED only after fitting.

Method	q (q/d)	Cluster sizes	ARI	NMI	ℓ	BIC	$\hat{\kappa}$
Spherical k-means	2000 (1.000)	1458/1406/1019	0.970	0.944	—	—	—
Sparse k-means	985 (0.492)	553/784/2546	0.137	0.313	—	—	—
Dense vMF shared κ	2000 (1.000)	1458/1406/1019	0.970	0.944	18,917,075	-37,784,564	732.2/732.2/732.2
Dense vMF free κ_k	2000 (1.000)	1462/1391/1030	0.976	0.954	18,921,337	-37,793,072	738.5/818.0/615.3
M-L	1942 (0.971)	1462/1405/1016	0.970	0.944	18,914,871	-37,804,965	730.0/730.0/730.0
M-CGL	1471 (0.736)	1456/1392/1035	0.976	0.954	18,919,480	-37,798,101	738.9/816.5/610.0
E-CGL	1420 (0.710)	1456/1392/1035	0.976	0.954	18,919,114	-37,798,212	738.3/815.8/610.4
E-ACGL	1438 (0.719)	1458/1391/1034	0.975	0.951	18,919,276	-37,798,237	738.0/815.9/611.4

Selected q refers to the method-specific target: weighted coordinates for sparse k -means, prototype-union coordinates for M-L, directional-contrast coordinates for M-CGL, and posterior-score contrast coordinates for E-CGL and E-ACGL. E-CGL and E-ACGL use one zero-augmented $L = 240$ KKT-motivated path with relative lower endpoint 10^{-4} ; M-CGL uses its $L = 240$ dense-score-proxy path, and M-L retains its Rossi-style event-driven threshold path. Likelihood and BIC are not defined for the two k -means procedures. BIC is used within method; cross-method values are descriptive because nominal-dimension rules differ (Section 3.3.3). $\hat{\kappa}$ is ordered as CISI/CRAN/MED after post-fit component alignment.

For coordinate j , define

$$R_j = \left(n^{-1} \sum_{i=1}^n x_{ij}^2 \right)^{1/2}, \quad I_{kj}^{(\text{ctr})} = R_j \left(\hat{\eta}_{kj} - \hat{\eta}_j \right), \quad I_j^{(\text{com})} = R_j \left| \hat{\eta}_j \right|.$$

After fitting, the E-CGL components were aligned to CISI, CRAN, and MED; their concentrations were 738.32, 815.85, and 610.42. Panel A of Fig. 4 selects the five largest positive root-mean-square (RMS)-scaled contributions for each matched component and displays their signed values $I_{kj}^{(\text{ctr})}$. CISI contrasts were led by *library*, *information*, and *librarian*; CRAN by *flow*, *mach*, and *pressure*; and MED by *tumor*, *inhibitor*, and *rat*. Panel B ranks inactive coordinates by $I_j^{(\text{com})}$ and shows those with the largest common RMS-scaled natural-parameter contributions. Their fitted values are equal across components, so their centered contrasts are numerically zero even though the common natural-parameter baseline remains in the fitted density.

5.4. Classic3 split stability

For each pair of the five prespecified Classic3 splits, the conditional support Jaccard index is the support-intersection size divided by the support-

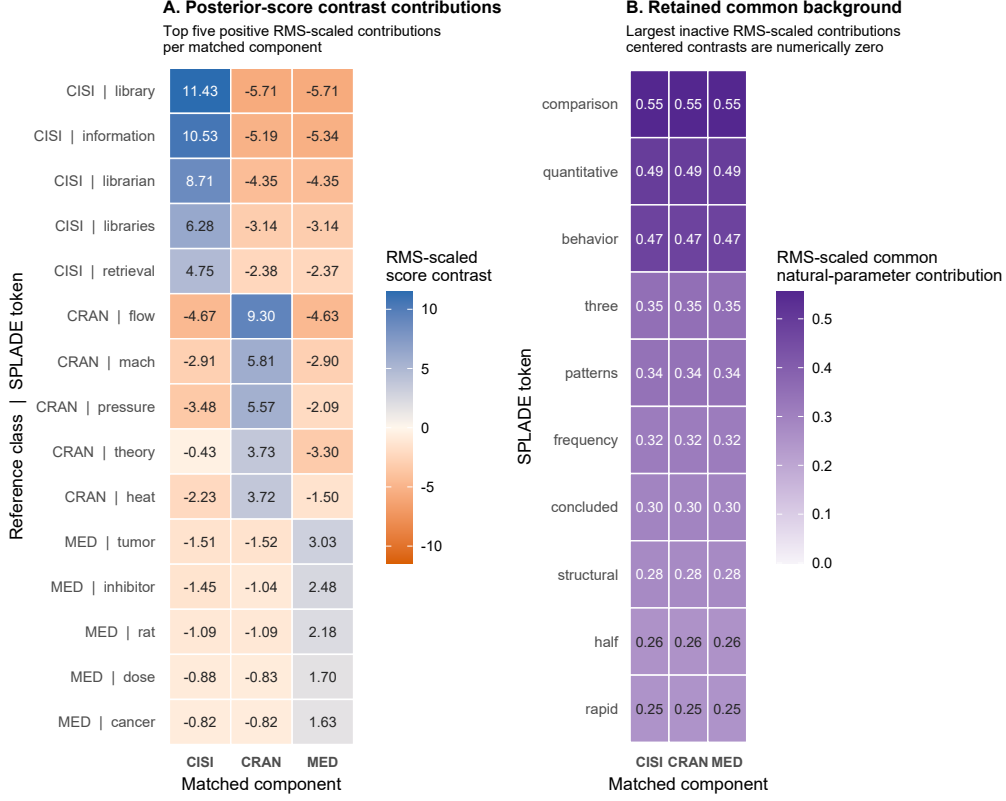


Figure 4: RMS-scaled posterior-score contrasts and common natural-parameter contributions in Classic3. E-CGL selected 1,420 of 2,000 coordinates. Panel A shows signed RMS-scaled posterior-score contrast contributions $I_{kj}^{(\text{ctr})}$ (blue, positive; orange, negative). Panel B shows the ten inactive coordinates with the largest RMS-scaled common contributions $I_j^{(\text{com})}$. Their equal component values represent retained common effects rather than posterior-score heterogeneity. Supplied labels were used only after fitting to align and name the components.

union size after both supports are restricted to the vocabulary shared by that pair. E-CGL attained a mean of 0.933 over the ten split pairs, and the 15 selected-contrast tokens displayed in Fig. 4 were retained in all five splits. Because the true coordinate support is unknown, support TPR, FPR, precision, and F_1 are not reported.

6. Conclusions

We introduced posterior-score contrast support for vMF mixtures and proposed E-CGL to estimate it by penalizing centered natural-parameter contrasts while preserving common baselines. This estimand identifies the coordinates entering the feature-dependent linear terms of pairwise posterior log-odds. It differs from prototype support and directional-contrast support, and the latter can diverge from posterior-score contrast support under concentration heterogeneity. The guarded proximal GEM path, target-preserving refit, and within-method BIC provide a practical computational procedure aligned with this target.

Across the evaluated $n = 1000$ primary-grid cells, E-CGL selected 16.02–16.41 coordinates on average and attained $F_{1,\eta}$ between 0.988 and 0.999 for a true support size of 16. At $n = 1000$, support recovery changed little across the evaluated oracle Bayes errors, although clustering agreement declined, showing that clustering and support difficulty need not move together. Recovery deteriorated when the same contrast signal was distributed over many weak coordinates; in the moderately dense $n = 300$ condition, $F_{1,\eta} = 0.805$ and ARI was 0.556.

In Classic3, E-CGL excluded 29.00% of coordinates while attaining ARI and NMI close to those of the dense component-specific-concentration fit. The five prespecified splits gave a mean conditional support Jaccard index of 0.933, and all 15 displayed contrast tokens were retained in every split. These findings describe stability under the fixed SPLADE representation and vocabulary filters, not recovery of an unknown population support or invariance to arbitrary rotations of the feature space.

The conclusions are conditional on fixed K and on the path-generated model collection. Refit BIC is a practical within-method criterion subject to the mixture singularity and local-optimization qualifications in Section 3.3.3; no general selection-consistency or global optimality claim is made. The theoretical results concern a fixed-dimensional conditional convex subproblem and an idealized residual-vanishing sequence, so the high-dimensional experiments provide computational rather than asymptotic evidence. Weak separation, diffuse signals, small samples, and local mixture optima can still cause underselection, while adaptive weighting remains sensitive to pilot contrasts and path coverage.

Further work includes effective degrees of freedom, joint uncertainty in K and support, and selection under diffuse weak signals. E-CGL is therefore in-

tended as a target-specific sparse mixture procedure rather than a uniformly superior replacement for dense or prototype-oriented models.

Appendix A. Proofs of theoretical results

The main text states the analytic bounds that determine each result; this appendix gives the complete arguments.

Proof of Lemma 1. The closed simplex and the product of the K closed Euclidean balls of radius $\kappa_{\max}$ are compact. Each vMF component density is positive and continuous on this set, so the observed log-likelihood is finite and continuous. The centered group penalty is also continuous. The criterion therefore attains its maximum by the extreme-value theorem. $\square$

Proof of Proposition 1. Taking the average of $v = \bar{v}\mathbf{1}_K + c$ and using $\mathbf{1}_K^\top c = 0$ determines $\bar{v}$ and hence c uniquely. The three zero-contrast statements are immediate. Finally, $\sum_{k < \ell} (v_k - v_\ell)^2 = K \sum_k (v_k - \bar{v})^2$, which gives the stated identity. $\square$

Proof of Proposition 2. Because $\kappa = \bar{\kappa}\mathbf{1}_K + h$ and $M = \mathbf{1}_K \bar{\mu}^\top + U$,

$$E = \text{diag}(\kappa)M = \bar{\kappa}\mathbf{1}_K \bar{\mu}^\top + \bar{\kappa}U + h\bar{\mu}^\top + \text{diag}(h)U.$$

Apply H_K , using $H_K\mathbf{1}_K = 0$, $H_KU = U$, and $H_Kh = h$. $\square$

Proof of Proposition 3. Because $P\mathbf{1}_K = \mathbf{1}_K$, P commutes with J_K and H_K . Permutation matrices are orthogonal and therefore preserve the Euclidean norm. Applying Proposition 4 gives equivariance of the proximal map. $\square$

Proof of Proposition 4. The spaces spanned by $\mathbf{1}_K$ and orthogonal to $\mathbf{1}_K$ are orthogonal. The penalty is zero on the first space and equals the Euclidean norm on the second. The minimization therefore preserves $J_K v$ and applies the standard group soft-thresholding map to $H_K v$. $\square$

Proof of Proposition 5. The vMF log-partition function has Hessian equal to the covariance matrix of a unit-norm random vector and is therefore positive semidefinite with operator norm at most one. The Hessian of the k th block of f_t consequently has operator norm at most N_k . The descent result follows by combining the smooth majorization inequality with optimality of the proximal update. $\square$

Proof of Theorem 1. For finite natural parameters, the vMF log-partition function is strictly convex. Its Hessian is the covariance matrix of a vMF random vector, and the vMF density is positive on the full sphere; hence $\text{Var}(a^\top X) > 0$ for every nonzero $a \in \mathbb{R}^d$. Since $N_k > 0$, the sum of its K blocks is strictly convex. The standard large-argument expansion of the modified Bessel function gives

$$-\log C_d(\kappa) = \kappa - \frac{d-1}{2} \log \kappa + R_d(\kappa), \quad \kappa \rightarrow \infty.$$

Since $R_d(\kappa) = O(1)$, there exist κ_0 and $B_d < \infty$ such that $R_d(\kappa) \geq -B_d$ for every $\kappa \geq \kappa_0$.

Consequently, as $\|\eta_k\|_2 \rightarrow \infty$,

$$\begin{aligned} N_k \{-\log C_d(\|\eta_k\|_2)\} - r_k^\top \eta_k &\geq (N_k - \|r_k\|_2) \|\eta_k\|_2 \\ &\quad - \frac{N_k(d-1)}{2} \log \|\eta_k\|_2 - N_k B_d. \end{aligned}$$

The positive linear coefficient implied by $\|r_k\|_2 < N_k$ dominates the logarithmic term. Thus f_t is coercive. Adding the finite convex group penalty preserves strict convexity and coercivity, so $F_t^{(a)}$ has a unique minimizer. If that minimizer is interior to the concentration bound, it is also the unique bounded-subproblem solution. The gradient of f_t is globally Lipschitz with constant no larger than $\max_k N_k$. Hence the initial trial step $1/\max_k N_k$ satisfies the exact majorization inequality and is bounded away from zero. Standard proximal-gradient theory yields convergence and the $O(u^{-1})$ objective rate (Beck and Teboulle, 2009). The strict inequality $\|r_k\|_2 < N_k$ excludes the degenerate case in which all observations receiving positive responsibility from a component have one common direction. The finite cap still gives existence in that case, but this theorem does not apply. $\square$

Proof of Proposition 6. The standard EM lower-bound identity expresses the observed log-likelihood difference as the auxiliary-function difference plus a nonnegative Kullback–Leibler divergence term. The penalty depends only on the parameters and can be subtracted from both sides, yielding the penalized result. $\square$

Proof of Theorem 2. Let $\Theta^{(t_j)} \rightarrow \bar{\Theta}$ be any convergent subsequence, which exists by compactness. Asymptotic regularity also gives $\Theta^{(t_j+1)} \rightarrow \bar{\Theta}$. After

passage to a further subsequence if needed, boundedness of the step sizes gives $s_{t_j} \rightarrow \bar{s}$ for some $\bar{s} \in [\underline{s}, \bar{s}]$. Exact posterior responsibilities are continuous on the assumed compact interior set, so $\tau^{(t_j)} \rightarrow \tau(\bar{\Theta})$.

The gradient $\nabla f_\tau(E)$ is jointly continuous in (E, τ) on this set. For s bounded away from zero, the scaled proximal map is jointly continuous in its input and in s . Moreover, $s_{t_j} \mathcal{G}_{s_{t_j}}(E^{(t_j+1)}; \tau^{(t_j)}) \rightarrow 0$. The vanishing natural-parameter residual therefore implies

$$\bar{E} = \text{prox}_{\bar{s}\lambda_\eta \mathcal{P}_a} \{\bar{E} - \bar{s} \nabla f_{\tau(\bar{\Theta})}(\bar{E})\}.$$

The proximal fixed-point characterization yields

$$0 \in \nabla f_{\tau(\bar{\Theta})}(\bar{E}) + \lambda_\eta \partial \mathcal{P}_a(\bar{E}).$$

At exact posterior responsibilities, Fisher’s identity for the observed-data score gives $\nabla f_{\tau(\bar{\Theta})}(\bar{E}) = -\nabla_E \ell(\bar{\Theta})$, proving the natural-parameter KKT inclusion. For the mixing block,

$$\pi^{(t_j+1)} - \frac{N(\tau^{(t_j)})}{n} \rightarrow 0,$$

while $\pi^{(t_j+1)} \rightarrow \bar{\pi}$ and $N(\tau^{(t_j)}) \rightarrow N\{\tau(\bar{\Theta})\}$. Hence $\bar{\pi}_k = N_k\{\tau(\bar{\Theta})\}/n$. Because the limiting mixing proportions are strictly positive,

$$\frac{\partial \ell(\bar{\Theta})}{\partial \pi_k} = \frac{N_k\{\tau(\bar{\Theta})\}}{\bar{\pi}_k} = n$$

for every k , which is the interior simplex KKT condition with $\nu = n$. The objective, residuals, and KKT system are equivariant under any fixed component permutation. The argument does not apply to boundary or finite-tolerance cases excluded by the assumptions (Green, 1990; Wu, 1983; Tseng, 2004). $\square$

Data and code availability

The source corpora must be obtained from their original providers under the applicable terms. Derived document-level SPLADE matrices are not redistributed because redistribution permission for the source corpora has not been established. A redistribution-safe reproducibility archive containing the reviewed implementation, tests, preprocessing specifications, configuration

files, random seeds and split records, exclusion records, vocabulary and payload checksums, compact numerical summaries, and result-generation scripts is available from the corresponding author on reasonable request. Large raw, path, cache, and private project files are not part of that archive.

Funding

This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT and Codex for English-language editing, translation, software-development assistance, and consistency checking. The authors reviewed and verified the resulting text, code, references, and numerical claims and take full responsibility for the content of the publication.

References

- Amayri, O., Bouguila, N., 2013. On online high-dimensional spherical data clustering and feature selection. *Engineering Applications of Artificial Intelligence* 26, 1386–1398. doi:10.1016/j.engappai.2012.10.009.
- Banerjee, A., Dhillon, I.S., Ghosh, J., Sra, S., 2005. Clustering on the unit hypersphere using von Mises–Fisher distributions. *Journal of Machine Learning Research* 6, 1345–1382. URL: <https://jmlr.org/papers/v6/banerjee05a.html>.
- Beck, A., Teboulle, M., 2009. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences* 2, 183–202. doi:10.1137/080716542.
- Belloni, A., Chernozhukov, V., 2013. Least squares after model selection in high-dimensional sparse models. *Bernoulli* 19, 521–547. doi:10.3150/11-BEJ410.

- Bondell, H.D., Reich, B.J., 2009. Simultaneous factor selection and collapsing levels in ANOVA. *Biometrics* 65, 169–177. doi:10.1111/j.1541-0420.2008.01061.x.
- Bouveyron, C., Brunet, C., 2012. Simultaneous model-based clustering and visualization in the Fisher discriminative subspace. *Statistics and Computing* 22, 301–324. doi:10.1007/s11222-011-9249-9.
- Bouveyron, C., Brunet-Saumard, C., 2014. Discriminative variable selection for clustering with the sparse Fisher-EM algorithm. *Computational Statistics* 29, 489–513. doi:10.1007/s00180-013-0433-6.
- Byrd, R.H., Lu, P., Nocedal, J., Zhu, C., 1995. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing* 16, 1190–1208. doi:10.1137/0916069.
- Chen, J., Chen, Z., 2008. Extended Bayesian information criteria for model selection with large model spaces. *Biometrika* 95, 759–771. doi:10.1093/biomet/asn034.
- Dempster, A.P., Laird, N.M., Rubin, D.B., 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)* 39, 1–22. doi:10.1111/j.2517-6161.1977.tb01600.x.
- Drton, M., Plummer, M., 2017. A Bayesian information criterion for singular models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 79, 323–380. doi:10.1111/rssb.12187.
- Eddelbuettel, D., François, R., 2011. Rcpp: Seamless R and C++ integration. *Journal of Statistical Software* 40, 1–18. doi:10.18637/jss.v040.i08.
- Formal, T., Lassance, C., Piwowarski, B., Clinchant, S., 2022. From distillation to hard negative sampling: Making sparse neural IR models more effective, in: *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, Association for Computing Machinery. pp. 2353–2359. doi:10.1145/3477495.3531857.
- Formal, T., Piwowarski, B., Clinchant, S., 2021. SPLADE: Sparse lexical and expansion model for first stage ranking, in: *Proceedings of the 44th*

- International ACM SIGIR Conference on Research and Development in Information Retrieval, Association for Computing Machinery. pp. 2288–2292. doi:10.1145/3404835.3463098.
- Gopal, S., Yang, Y., 2014. Von Mises–Fisher clustering models, in: Proceedings of the 31st International Conference on Machine Learning, PMLR. pp. 154–162. URL: <https://proceedings.mlr.press/v32/gopal14.html>.
- Green, P.J., 1990. On use of the EM algorithm for penalized likelihood estimation. *Journal of the Royal Statistical Society: Series B (Methodological)* 52, 443–452. doi:10.1111/j.2517-6161.1990.tb01798.x.
- Guo, J., Levina, E., Michailidis, G., Zhu, J., 2010. Pairwise variable selection for high-dimensional model-based clustering. *Biometrics* 66, 793–804. doi:10.1111/j.1541-0420.2009.01341.x.
- Hubert, L., Arabie, P., 1985. Comparing partitions. *Journal of Classification* 2, 193–218. doi:10.1007/BF01908075.
- Meinshausen, N., 2007. Relaxed lasso. *Computational Statistics & Data Analysis* 52, 374–393. doi:10.1016/j.csda.2006.12.019.
- Meynet, C., 2012. Variable Selection in Model-Based Clustering for High-Dimensional Data. Ph.D. thesis. Université Paris-Sud. URL: https://www.imo.universite-paris-saclay.fr/select/phd/VD_MEYNET_CAROLINE_09112012.pdf.
- Meynet, C., Maugis-Rabusseau, C., 2012. A sparse variable selection procedure in model-based clustering. URL: <https://hal.science/hal-00734316>. preprint, HAL hal-00734316.
- Ng, T.L.J., 2023. Penalized maximum likelihood estimator for mixture of von Mises–Fisher distributions. *Metrika* 86, 181–203. doi:10.1007/s00184-022-00867-0.
- Pan, W., Shen, X., 2007. Penalized model-based clustering with application to variable selection. *Journal of Machine Learning Research* 8, 1145–1164. URL: <https://www.jmlr.org/papers/v8/pan07a.html>.
- Parikh, N., Boyd, S., 2014. Proximal algorithms. *Foundations and Trends in Optimization* 1, 127–239. doi:10.1561/24000000003.

- Rossi, F., Barbaro, F., 2022. Mixture of von Mises–Fisher distribution with sparse prototypes. *Neurocomputing* 501, 41–74. doi:10.1016/j.neucom.2022.05.118.
- Salah, A., Nadif, M., 2019. Directional co-clustering. *Advances in Data Analysis and Classification* 13, 591–620. doi:10.1007/s11634-018-0323-4.
- Salah, A., Rogovschi, N., Nadif, M., 2016. Model-based co-clustering for high dimensional sparse data, in: *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, PMLR. pp. 866–874. URL: <https://proceedings.mlr.press/v51/salah16.html>.
- Schwarz, G., 1978. Estimating the dimension of a model. *The Annals of Statistics* 6, 461–464. doi:10.1214/aos/1176344136.
- Strehl, A., Ghosh, J., 2002. Cluster ensembles—a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research* 3, 583–617. URL: <https://jmlr.org/papers/v3/strehl02a.html>.
- Taghia, J., Ma, Z., Leijon, A., 2014. Bayesian estimation of the von Mises–Fisher mixture model with variational inference. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 36, 1701–1715. doi:10.1109/TPAMI.2014.2306426.
- Tseng, P., 2004. An analysis of the EM algorithm and entropy-like proximal point methods. *Mathematics of Operations Research* 29, 27–44. doi:10.1287/moor.1030.0073.
- Wang, H., Leng, C., 2008. A note on adaptive group lasso. *Computational Statistics & Data Analysis* 52, 5277–5286. doi:10.1016/j.csda.2008.05.006.
- Witten, D.M., Tibshirani, R., 2010. A framework for feature selection in clustering. *Journal of the American Statistical Association* 105, 713–726. doi:10.1198/jasa.2010.tm09415.
- Wu, C.F.J., 1983. On the convergence properties of the EM algorithm. *The Annals of Statistics* 11, 95–103. doi:10.1214/aos/1176346060.
- Xie, B., Pan, W., Shen, X., 2008. Variable selection in penalized model-based clustering via regularization on grouped parameters. *Biometrics* 64, 921–930. doi:10.1111/j.1541-0420.2007.00955.x.

- Yuan, M., Lin, Y., 2006. Model selection and estimation in regression with grouped variables. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 68, 49–67. doi:10.1111/j.1467-9868.2005.00532.x.
- Zou, H., 2006. The adaptive lasso and its oracle properties. *Journal of the American Statistical Association* 101, 1418–1429. doi:10.1198/016214506000000735.